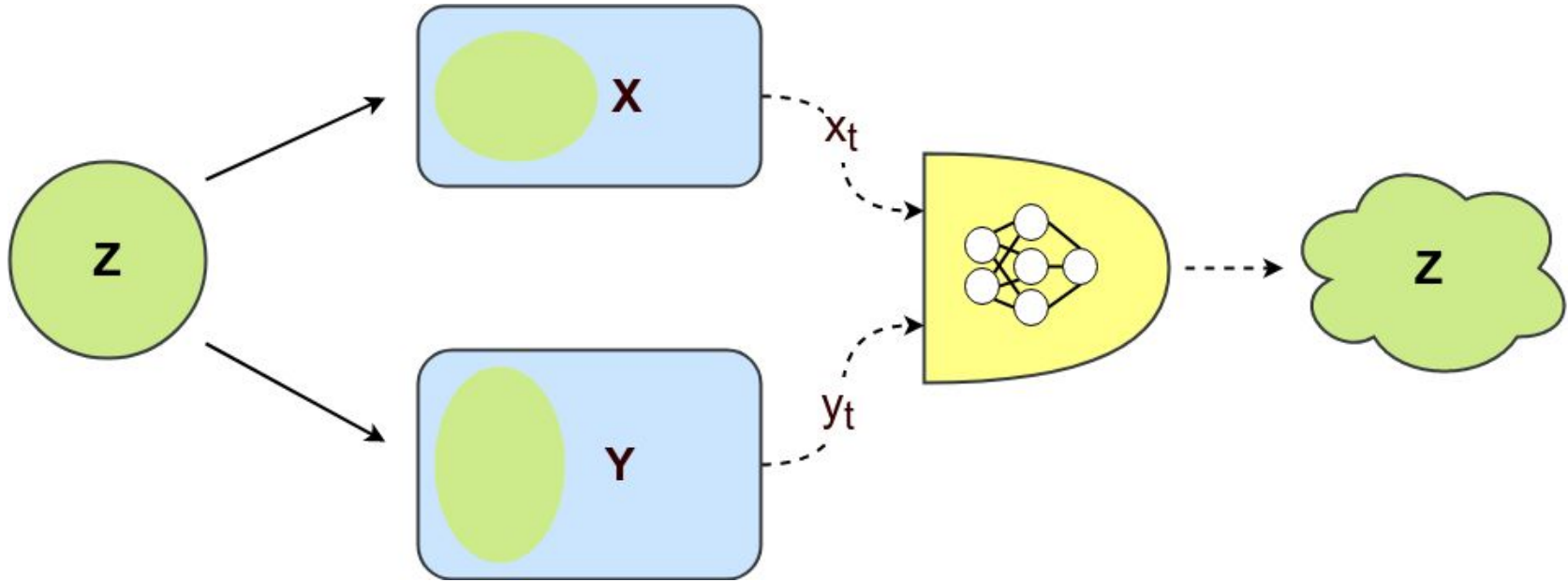


Hidden Common Driver Reconstruction and the Anisotropic Self-Organizing Map

Zsigmond Benko
BCNF
11/14/2025

Shared driver reconstruction with SOM

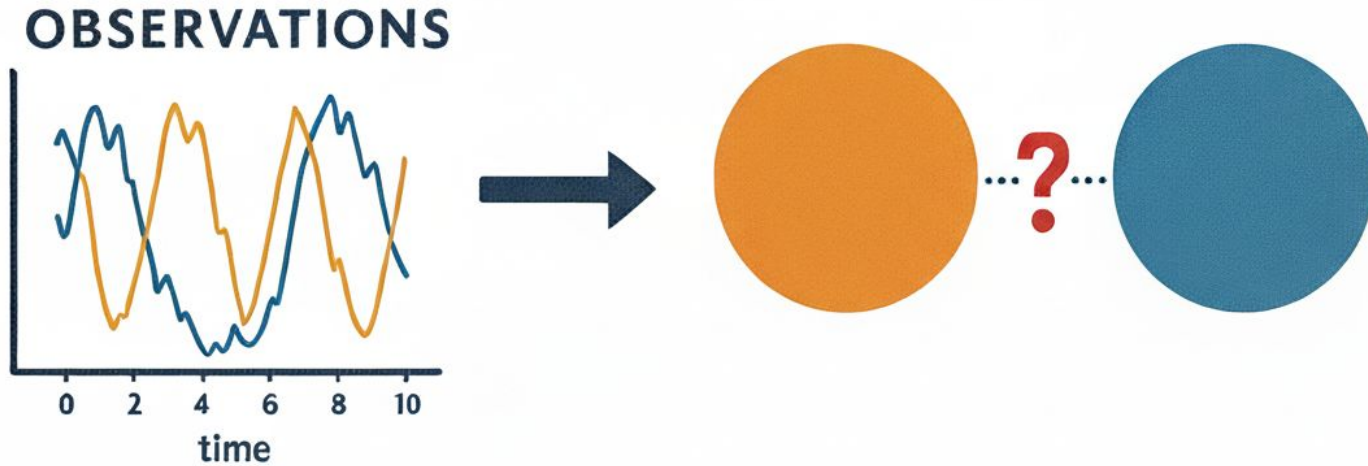




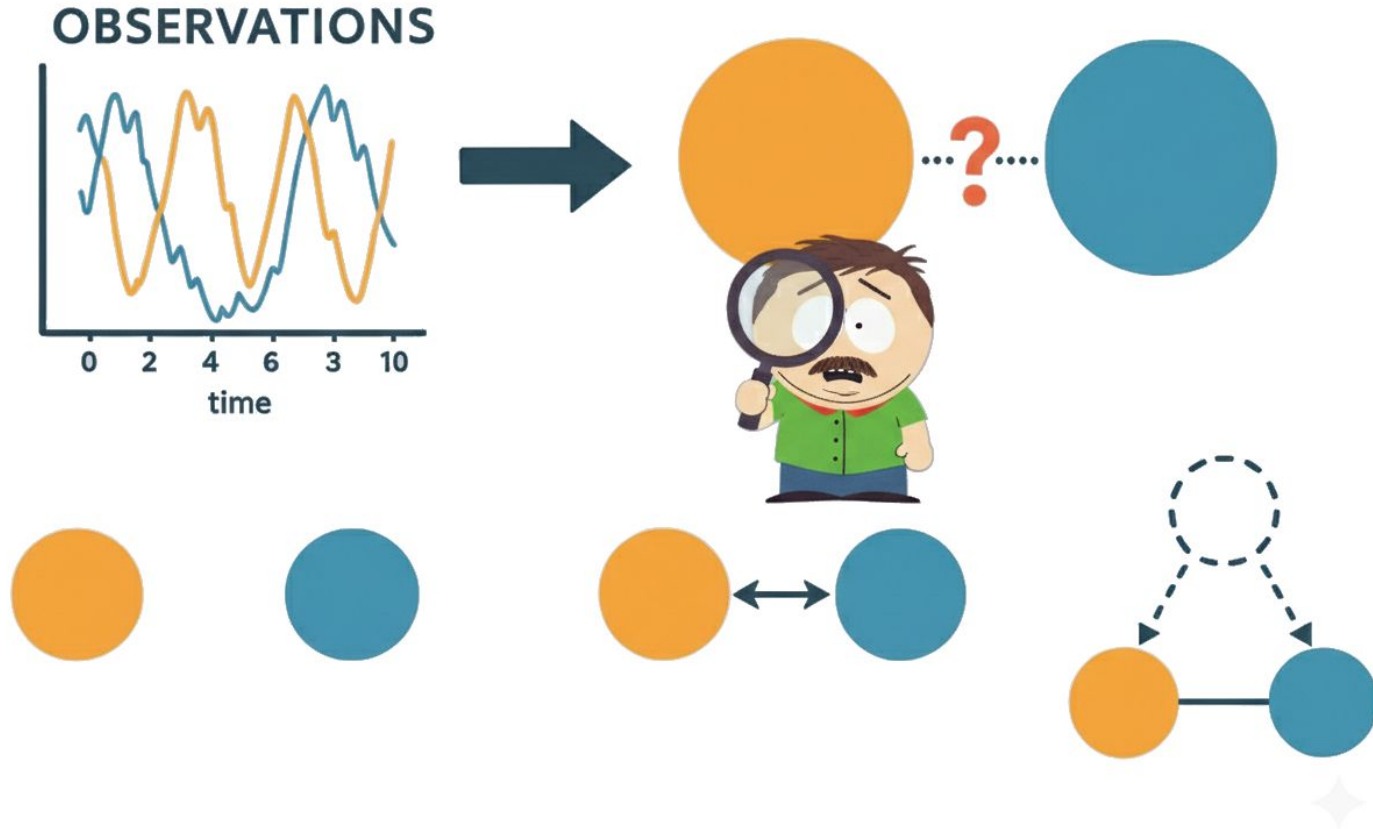
**BUT HOW TO TELL IF
THERE WAS HIDDEN
COMMON CAUSE?**



Causal Discovery from time series



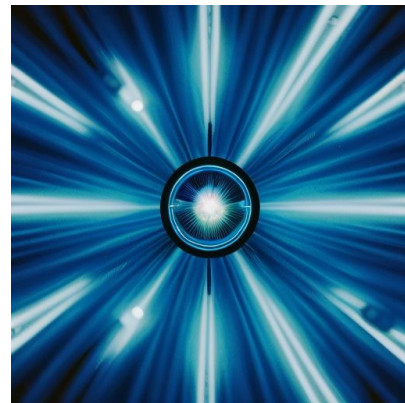
Causal Discovery from time series



Causal Discovery Algorithms

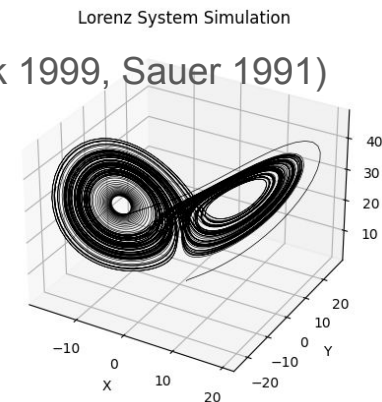
- Information Theory & Probability

- Statistics and Information theory (Wiener 1956)
- Granger causality (Granger 1969) -
- Transfer Entropy (Schreiber 2000)
- Geweke Granger Causality (1982), Transfer Entropy Spectrum (Tian 2022)
- Coupling in complex systems as information transfer across time scales (Palus 2019)



- Nonlinear Dynamics

- Theory of nonlinear deterministic dynamical systems (Takens 1971, Stark 1999, Sauer 1991)
- Convergent Cross Mapping (Sugihara 2012)
- Dimensional Causality (Stippinger 2018)
- Frequency-Domain Convergent Cross Mapping (Avvaru 2023)
- Cross-frequency Symbolic Cross Mapping (Gu 2023)



Theoretical interlude 1 – Dynamics and Causality



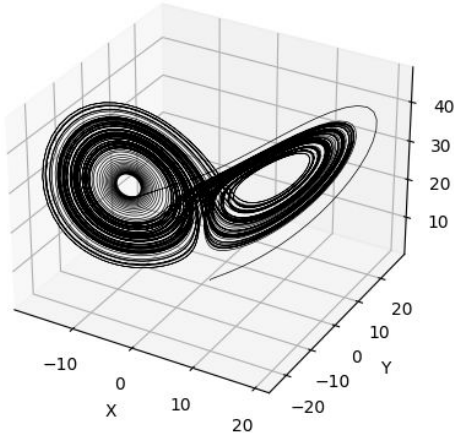
Nonlinear dynamical systems

Continuous time dynamics

$$\dot{s} = \varphi(s)$$

- Lorenz system

Lorenz System Simulation



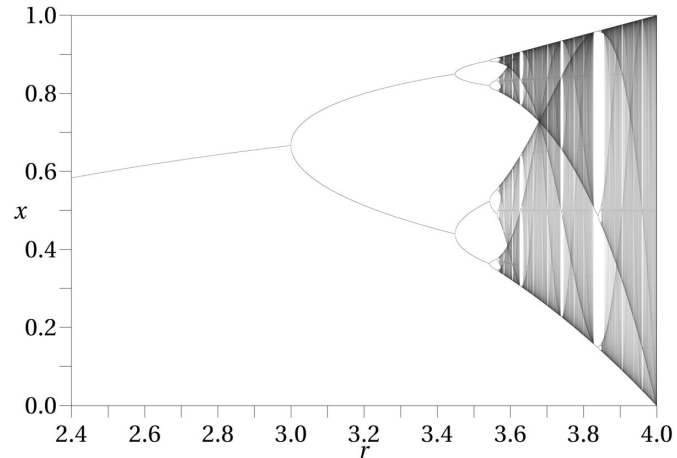
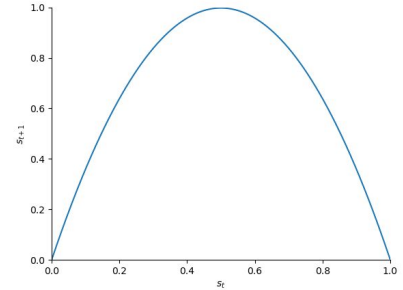
$$\begin{aligned}\frac{dx}{dt} &= \sigma(y - x) \\ \frac{dy}{dt} &= x(\rho - z) - y \\ \frac{dz}{dt} &= xy - \beta z\end{aligned}$$

Discrete-time dynamics

$$s_{t+1} = f(s_t)$$

- Logistic map

$$x_{t+1} = rx_t(1 - x_t)$$



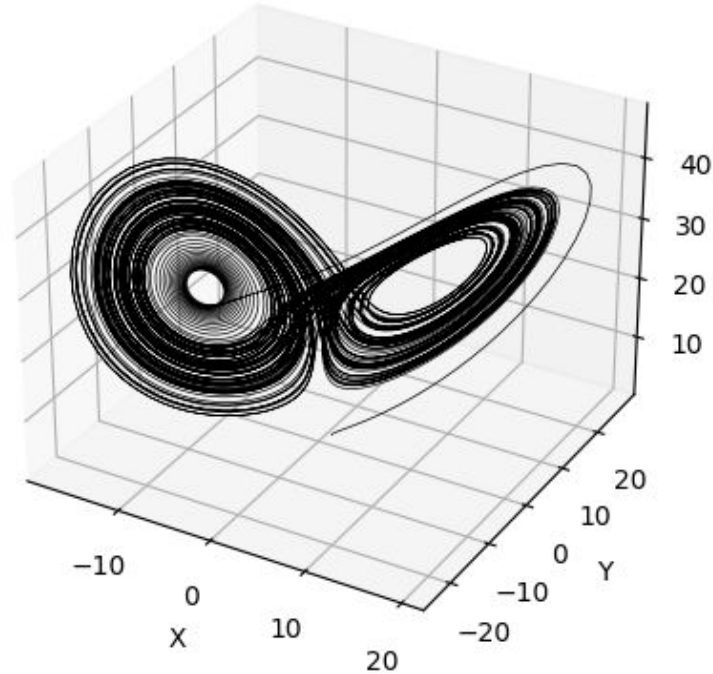
Time Delay Embedding (Takens 1971)

$$\frac{dx}{dt} = \sigma(y - x)$$

$$\frac{dy}{dt} = x(\rho - z) - y$$

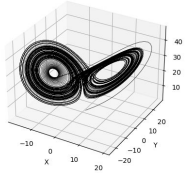
$$\frac{dz}{dt} = xy - \beta z$$

Lorenz System Simulation



Lorenz System Simulation

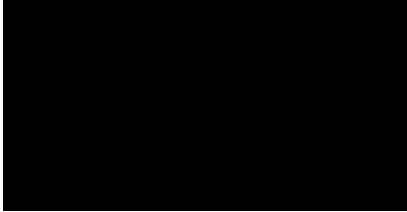
$$\begin{aligned}\frac{dx}{dt} &= \sigma(y - x) \\ \frac{dy}{dt} &= x(\rho - z) - y \\ \frac{dz}{dt} &= xy - \beta z\end{aligned}$$



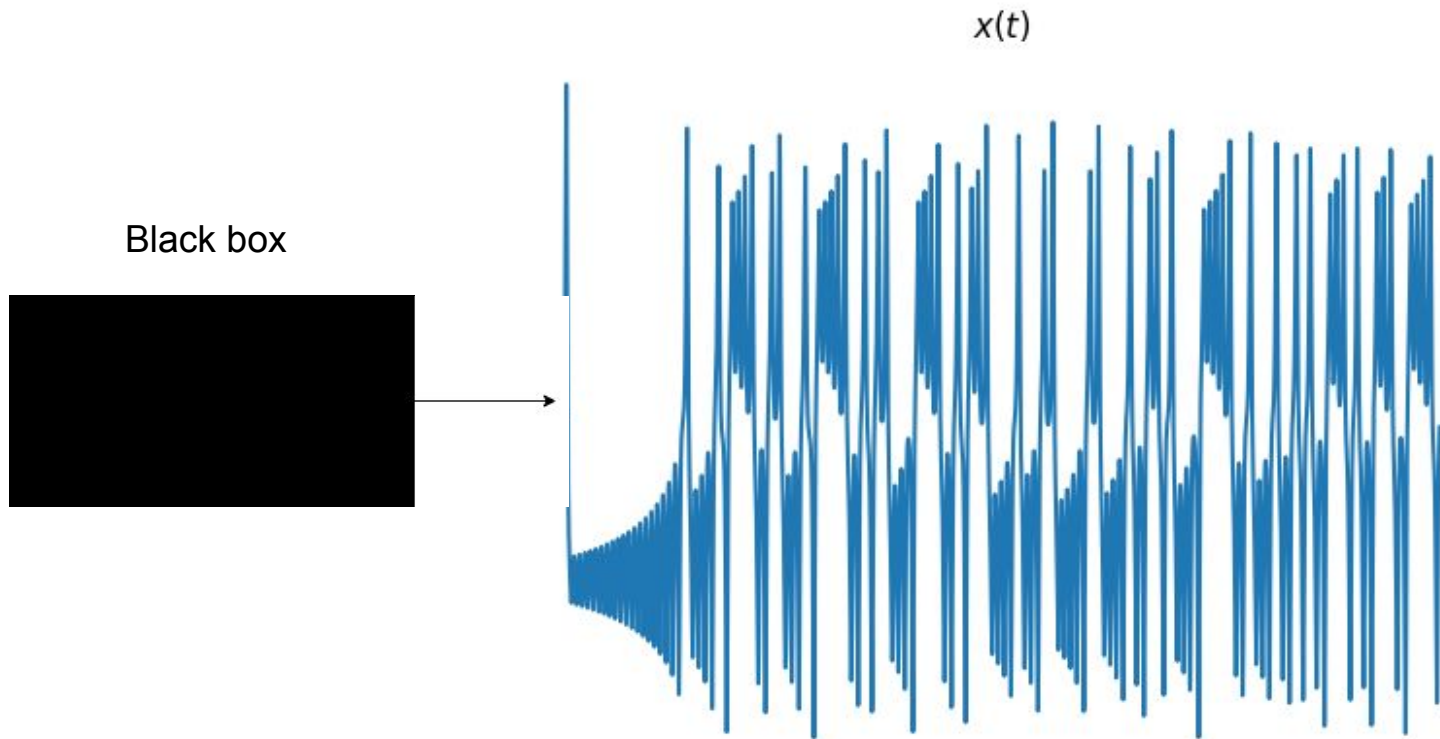
Time Delay Embedding (Takens 1971)

State and governing equations are unknowns

Black box



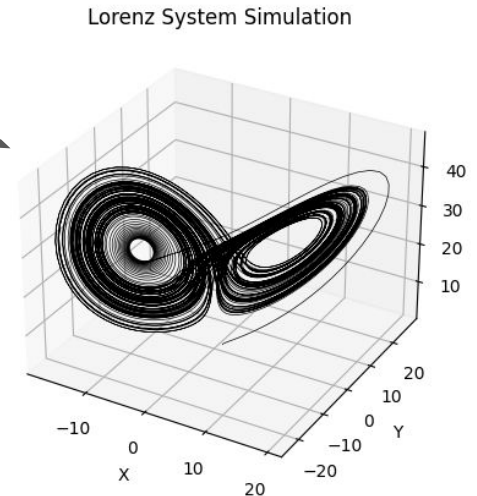
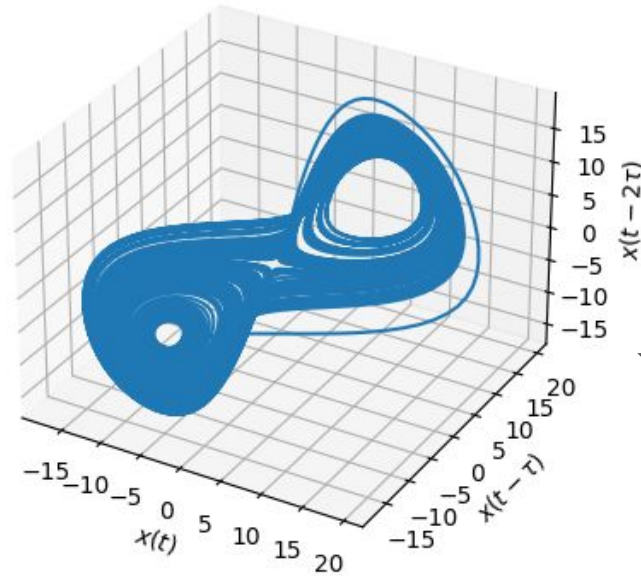
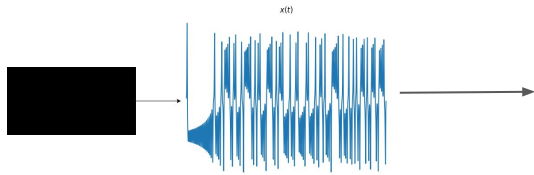
Time Delay Embedding (Takens 1971)



Time Delay Embedding (Takens 1971)

$$X(t) = [x(t), x(t - \tau), x(t - 2\tau)]$$

The state can be reconstructed!

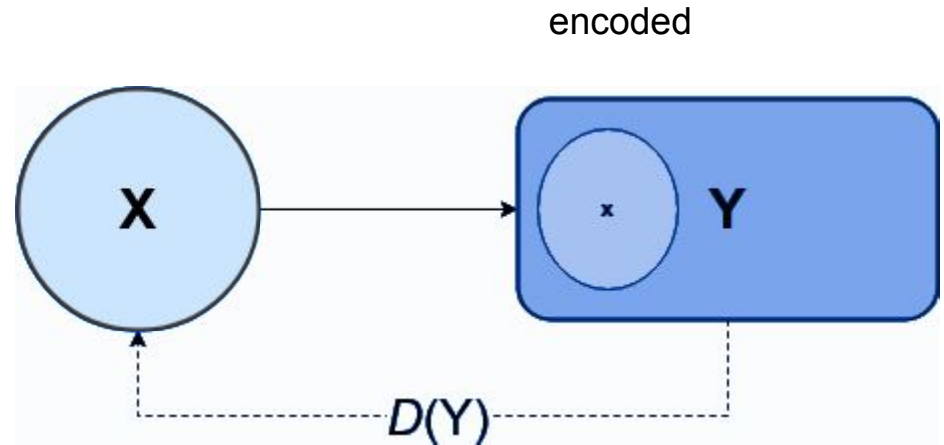


Takens, F. (1981). Detecting strange attractors in turbulence. *Dynamical Systems and Turbulence, Warwick 1980*, 898, 366–381. <https://doi.org/10.1007/BFb0091924>

Packard, N. H., Crutchfield, J. P., Farmer, J. D., & Shaw, R. S. (1980). Geometry from a Time Series. *Physical Review Letters*, 45(9), 712–716. <https://doi.org/10.1103/PhysRevLett.45.712>

Nonlinear Dynamical Systems & Causality (Stark 1999)

$$\dot{X} = f(X)$$
$$\dot{Y} = g(Y, X)$$



Mapping exists

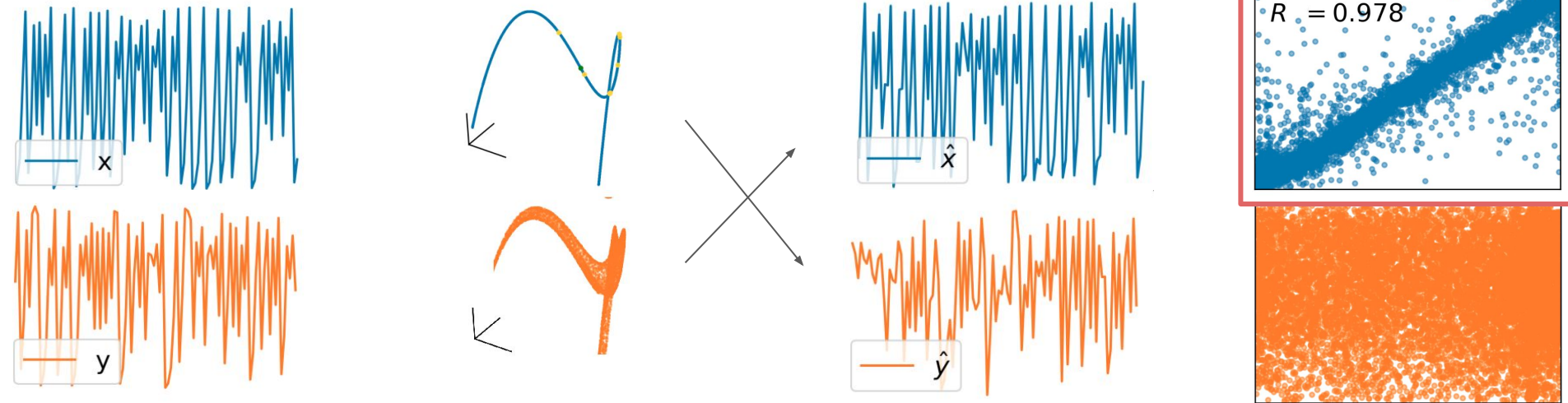
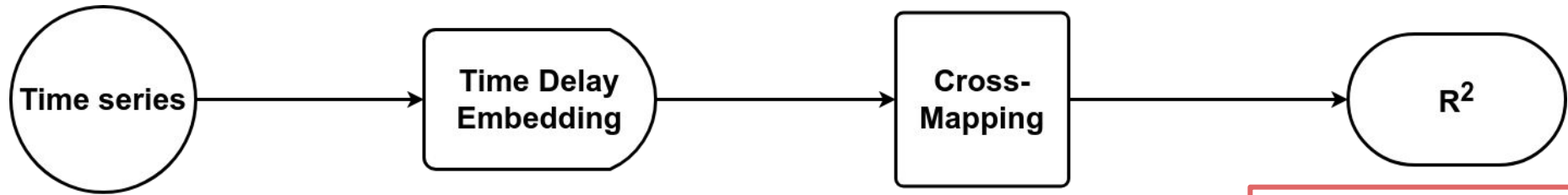
Methods 1 – Towards detecting Hidden Common Driver
and measuring its dimension

Convergent **Cross-Mapping** (Sugihara et al, 2012)

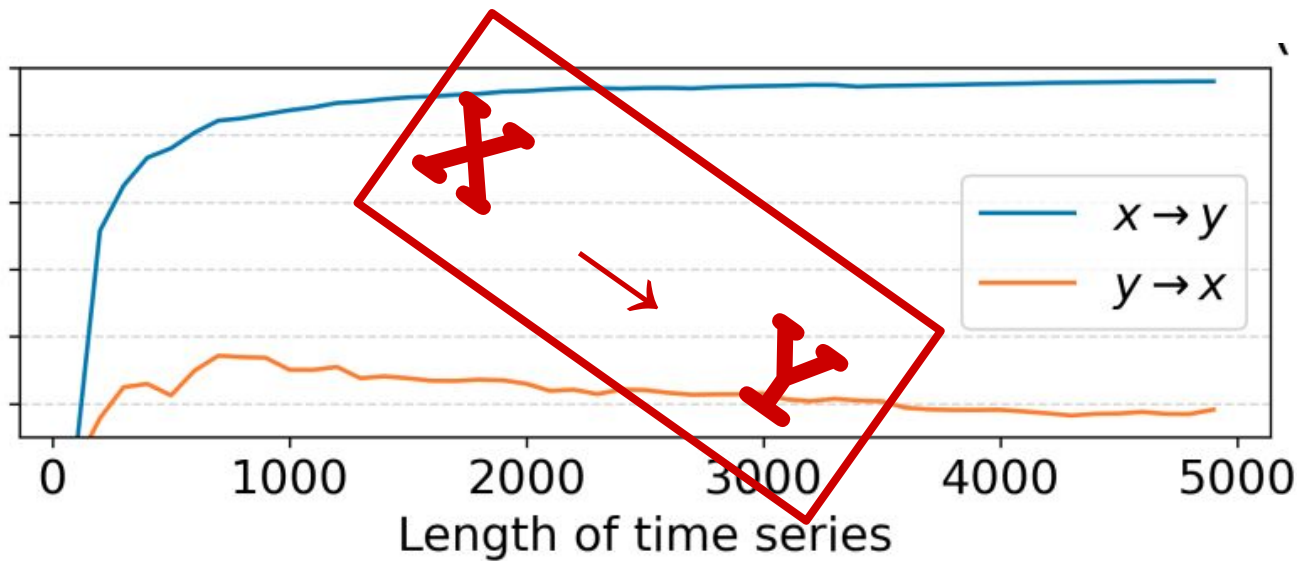
1. Embedding

2. Prediction

3. Evaluation



Convergent Cross-Mapping



Effect Time Delay



Article | [Open access](#) | Published: 26 March 2019

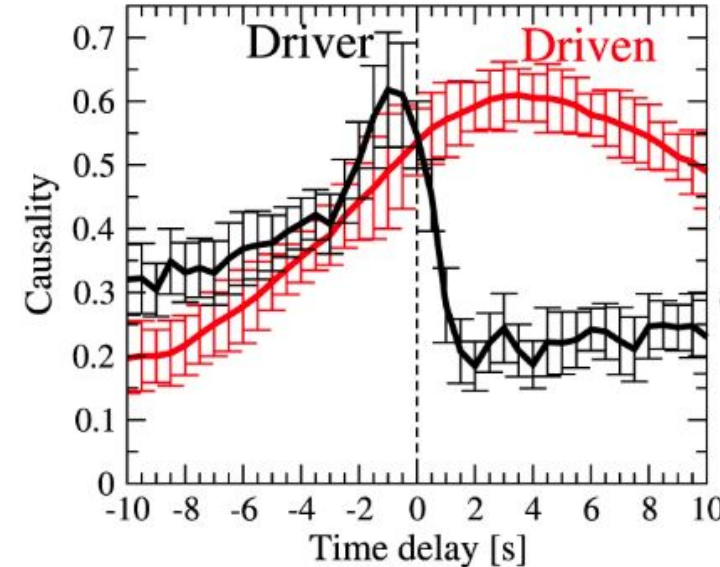
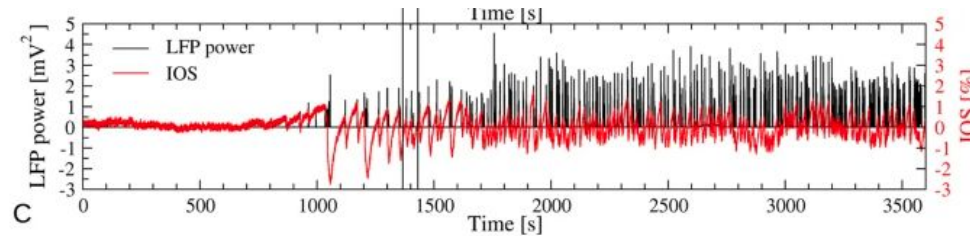
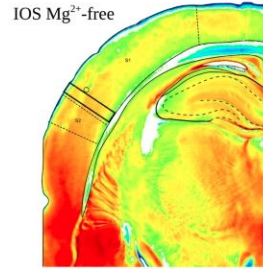
Causal relationship between local field potential and intrinsic optical signal in epileptiform activity *in vitro*

[Zsigmond Benkő](#), [Kinga Moldován](#), [Katalin Szádeczky-Kardoss](#), [László Zalányi](#), [Sándor Borbély](#), [Ildikó Világi](#)

& [Zoltán Somogyvári](#) ✉

[Scientific Reports](#) 9, Article number: 5171 (2019) | [Cite this article](#)

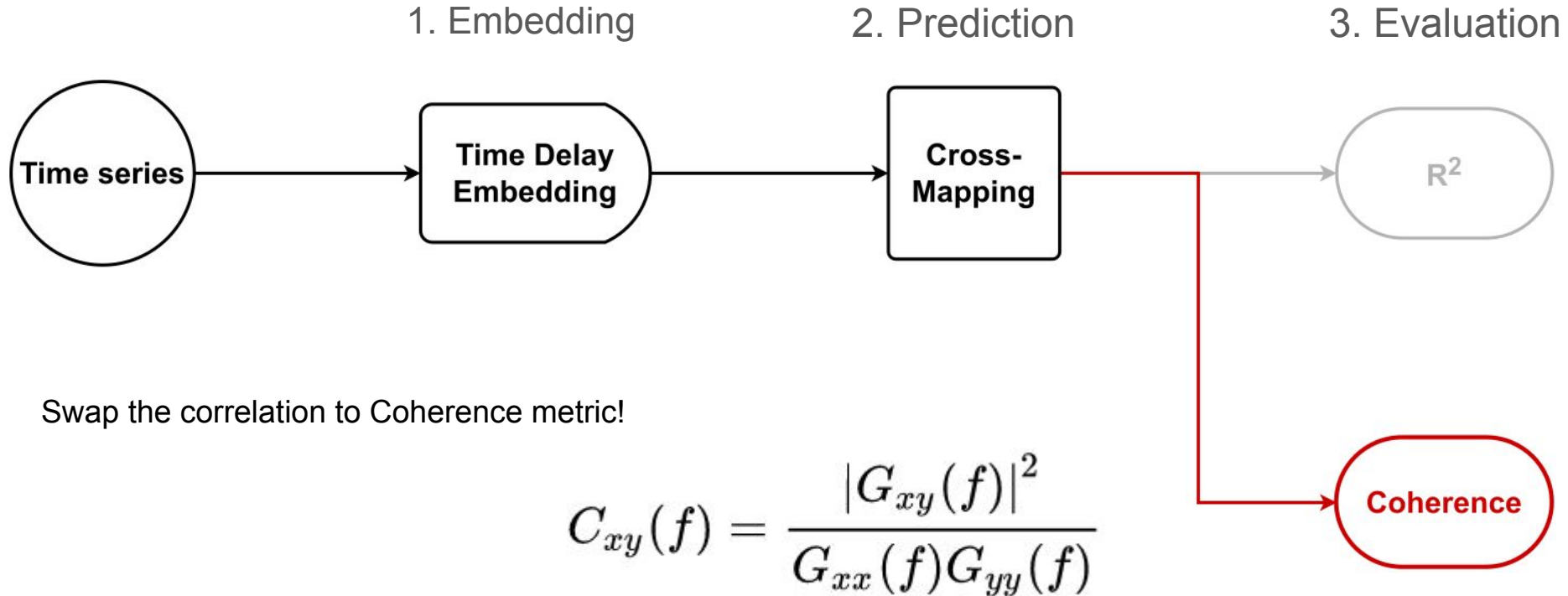
- Shifting the time-series relatively in time (Ye et al, 2015)
- Application to LFP vs. iOS (Benko et al 2019)



Ye, Hao, Ethan R. Deyle, Luis J. Gilarranz, and George Sugihara. “Distinguishing Time-Delayed Causal Interactions Using Convergent Cross Mapping.” *Scientific Reports* 5, no. 1 (2015): 14750. <https://doi.org/10.1038/srep14750>.

Benkő, Zsigmond, Kinga Moldován, Katalin Szádeczky-Kardoss, et al. “Causal Relationship between Local Field Potential and Intrinsic Optical Signal in Epileptiform Activity in Vitro.” *Scientific Reports* 9, no. 1 (2019): 5171. <https://doi.org/10.1038/s41598-019-41554-x>.

Cross-Mapping Coherence



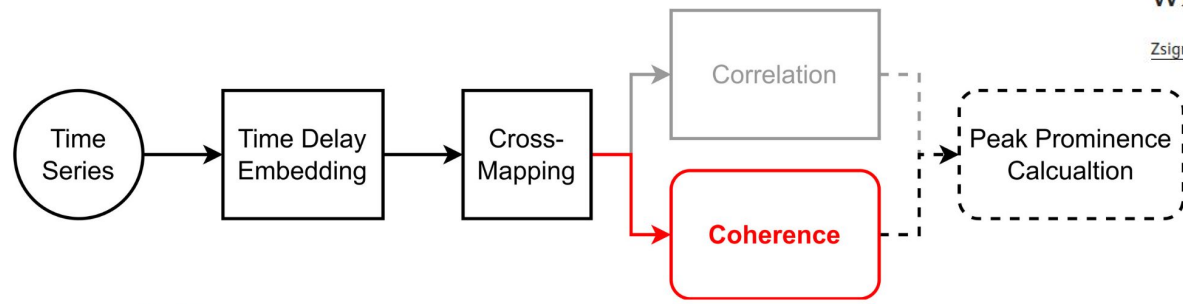


Cross-Mapping Coherence

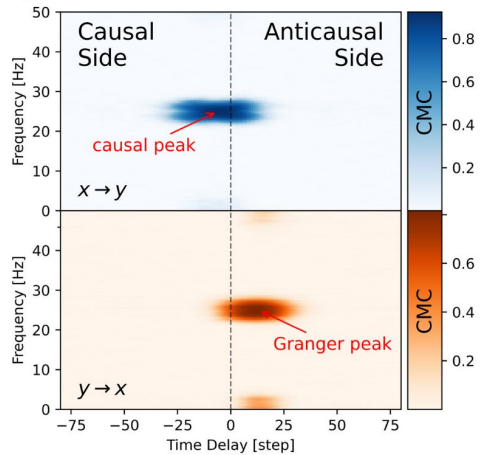
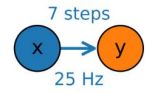
Detecting causality in the frequency domain with Cross-Mapping Coherence

Zsigmond Benkő^a, Bálint Varga^{a, b}, Marcell Stippinger^a, Zoltán Somogyvári^a

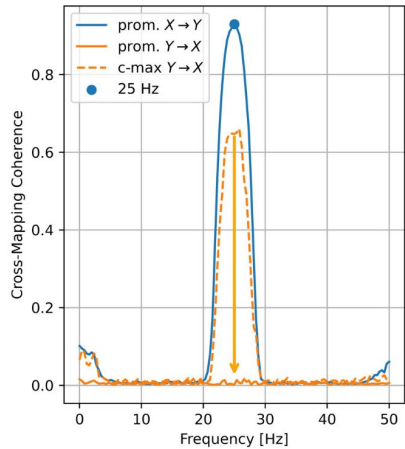
(a)



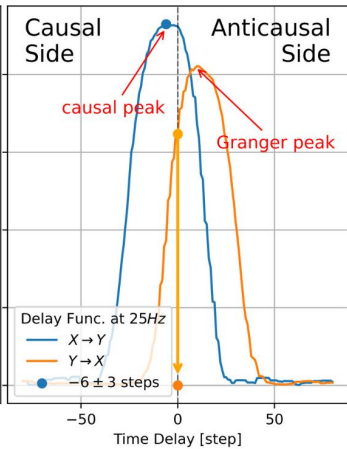
(b)



(c)



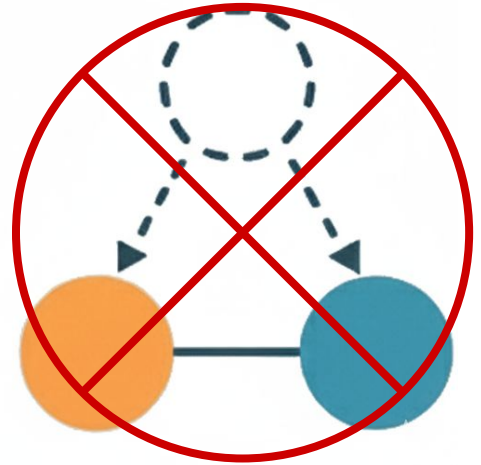
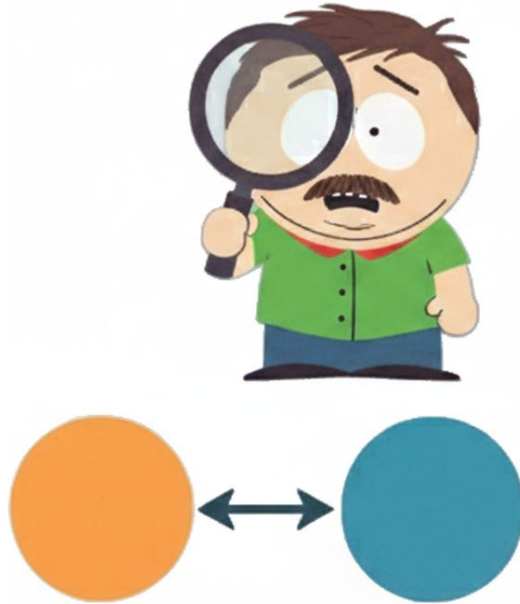
(d)



- Frequency-Domain causal information
- Disentangle predictability from causality with peak prominence calculation

Not for Common cause

- CCM detects directional or circular couplings
- But not hidden common drivers!

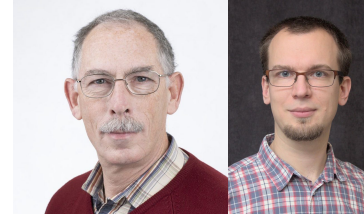


Detecting Hidden Common Driver

- Recurrence plot-based method (Hirata et al, 2010)
- Robust methods (Sauer & Sugihara, 2025)
- Dimensionality-based Causal inference
 - Relation between causality and manifold dimensions (Wojcik, 2001)
 - Full Bayesian implementation of the principles (Benko et al, 2024)

Bayesian inference of causal relations between dynamical systems

Zsigmond Benkő ^{a,1}, Ádám Zlatniczki ^{a,b,h,1}, Marcell Stippinger ^{a,1}, Dániel Fabó ^{c,d},
 András Sólyom ^c, Loránd Erőss ^e, András Telcs ^{a,f,2}, Zoltán Somogyvári ^{a,g,2}  

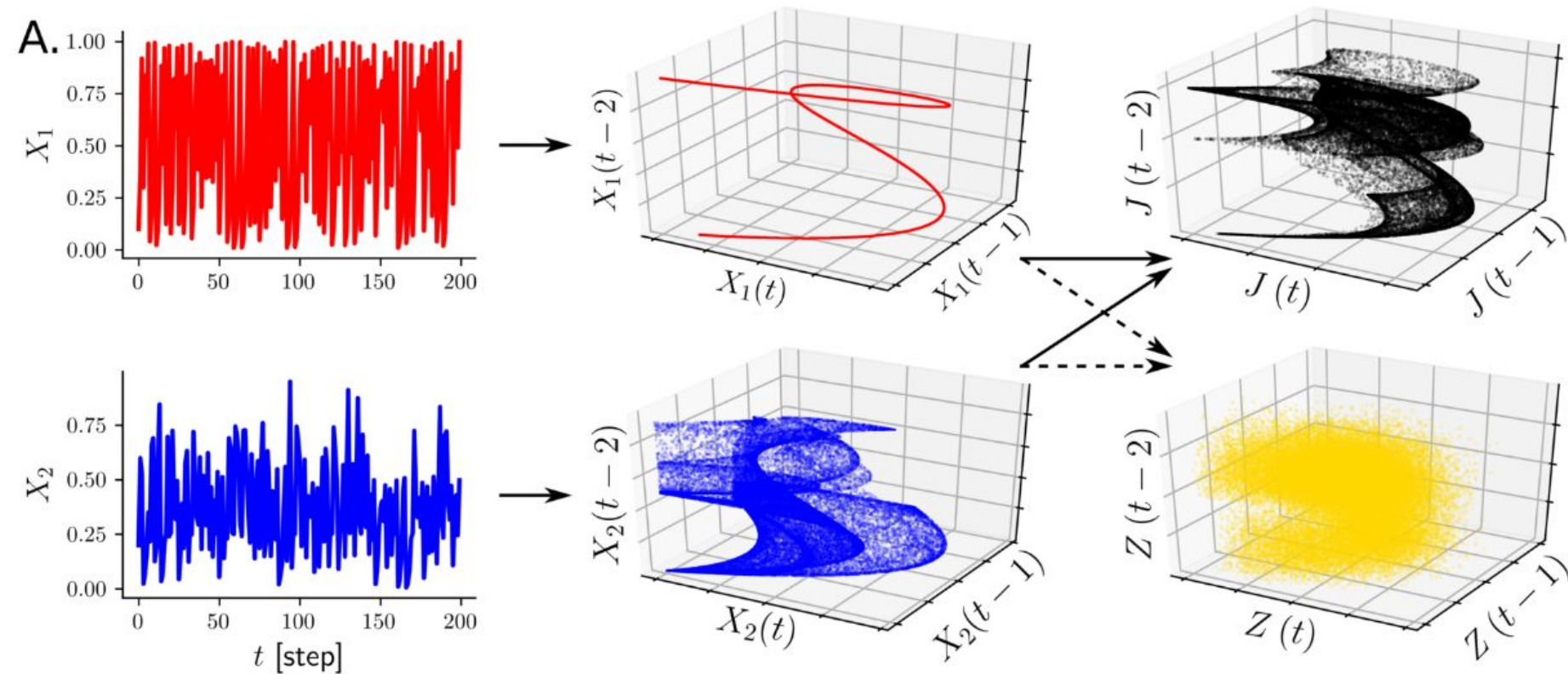


Hirata, Yoshito, and Kazuyuki Aihara. "Identifying Hidden Common Causes from Bivariate Time Series: A Method Using Recurrence Plots." *Physical Review E* 81, no. 1 (2010): 016203. <https://doi.org/10.1103/PhysRevE.81.016203>.

Sauer, Timothy, and George Sugihara. "Robust Methods to Detect Coupling among Nonlinear Dynamic Time Series." arXiv:2412.11252. Preprint, arXiv, May 6, 2025. <https://doi.org/10.48550/arXiv.2412.11252>.

Wójcik, Daniel, Andrzej Nowak, and Marek Kuś. "Dimension of Interaction Dynamics." *Physical Review E* 63, no. 3 (2001): 036221. <https://doi.org/10.1103/PhysRevE.63.036221>.

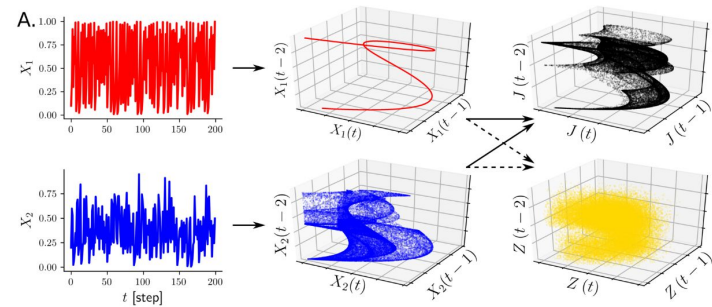
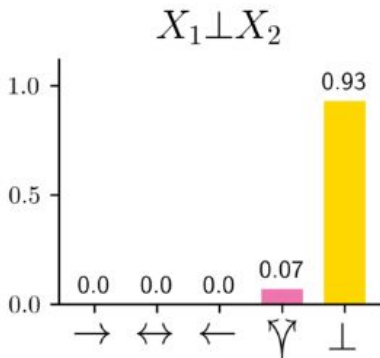
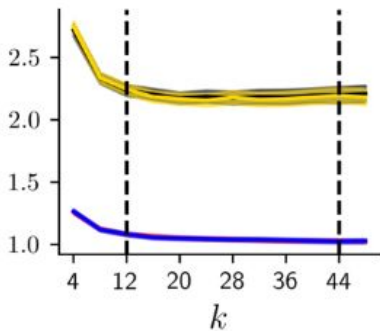
Dimensional Causality



Dimensions & Causality

$$X_1 \perp X_2$$

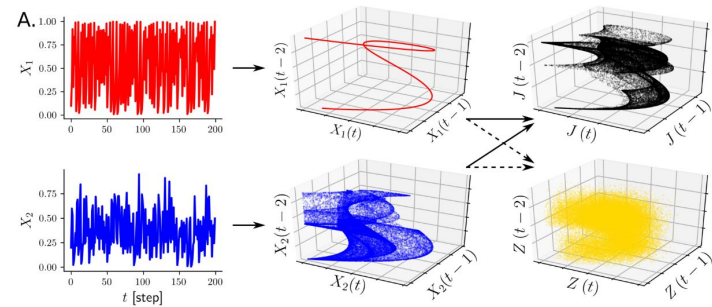
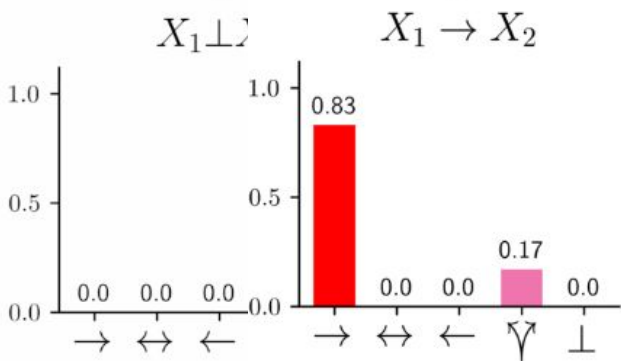
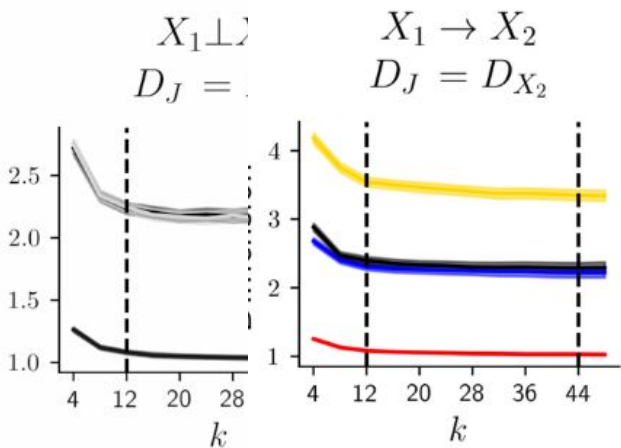
$$D_J = D_Z$$



Independent case

$$X \perp Y \iff D_X + D_Y = D_J$$

Dimensions & Causality

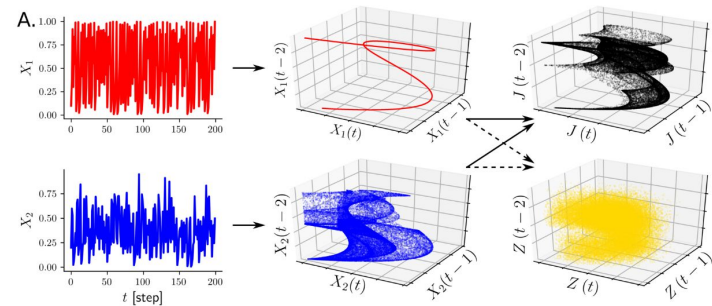
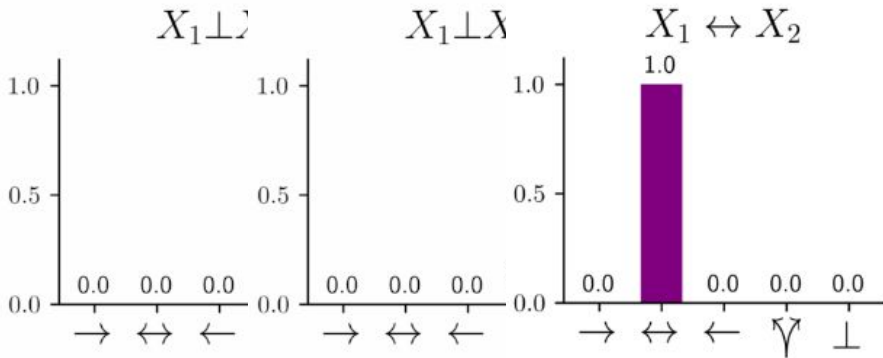
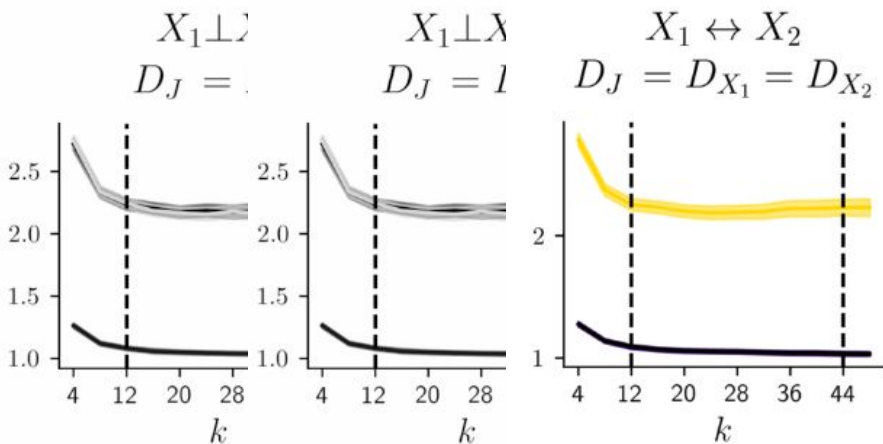


Unidirectional cases

$$X \rightarrow Y \iff D_X < D_Y = D_J$$

$$X \leftarrow Y \iff D_Y < D_X = D_J$$

Dimensions & Causality

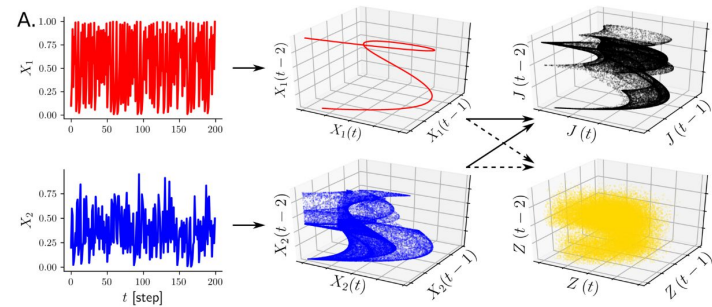
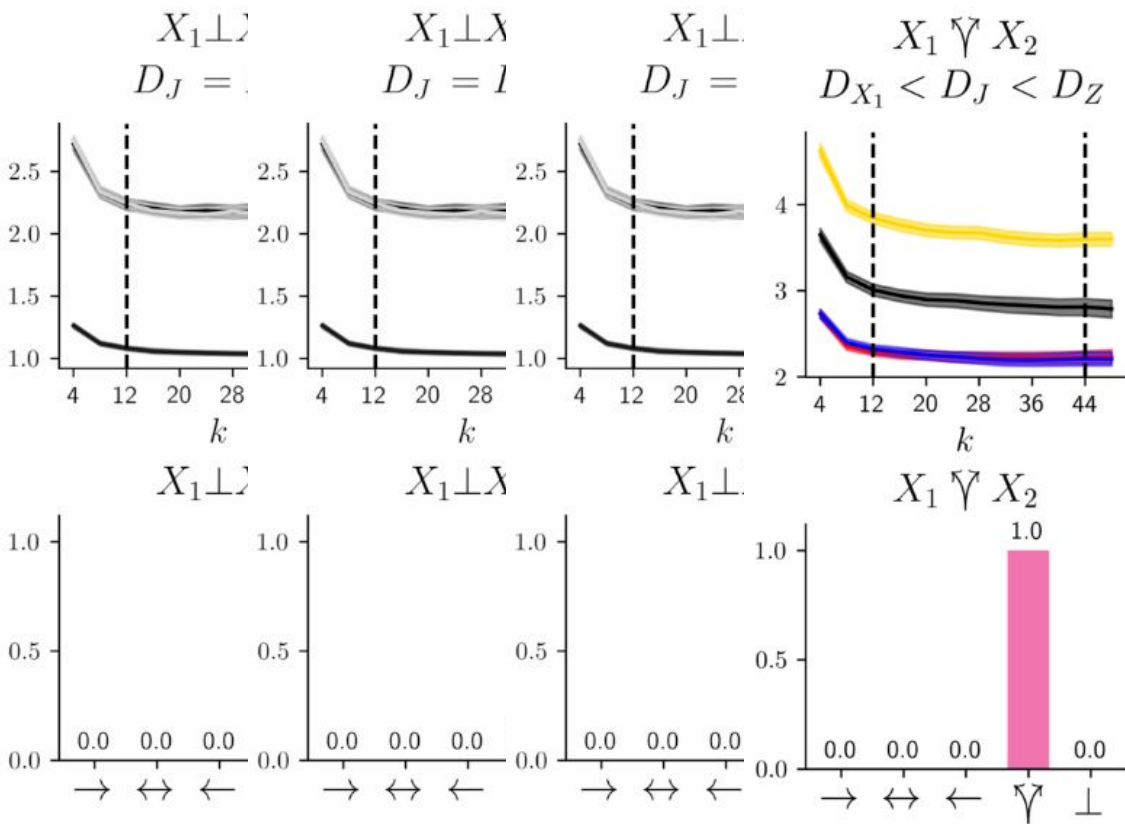


Bidirectional case

$$X \leftrightarrow Y \iff D_X = D_Y = D_J$$



Dimensions & Causality

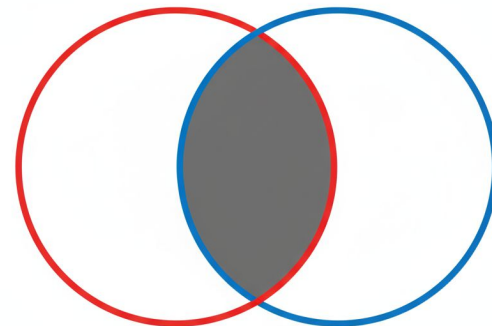
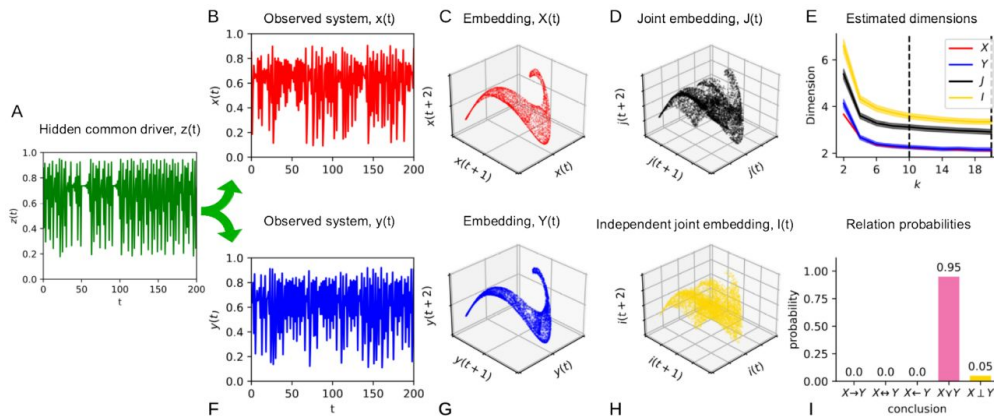


Common cause

$$X \nearrow Y \iff \max(D_X, D_Y) < D_J < D_X + D_Y$$

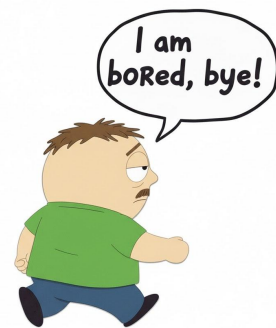


We can measure the dimension of the driver!



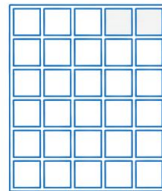
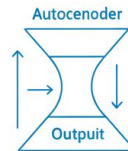
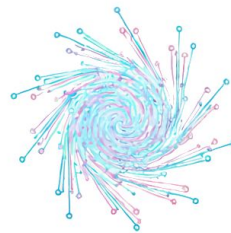
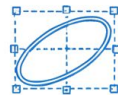
$$D_{HCC} = D_X + D_Y - D_J$$

Shared Dynamics Reconstruction



Dimensionality reduction methods & shared driver reconstruction

- Linear methods:
 - PCA, ICA, DCA, Canonical Correlation Analysis
- Non-linear extensions:
 - k-PCA, k-ICA, Deep Canonical Correlation Analysis
- Slow Feature Analysis (Wiskott, 2003)
- Dynamical Systems approach
 - Shared driver algorithm (Sauer, 2004)
 - ShRec (Gilpin, 2024)
- VAE-based (Yan et al., 2024)
- Self organizing Map (SOM)?



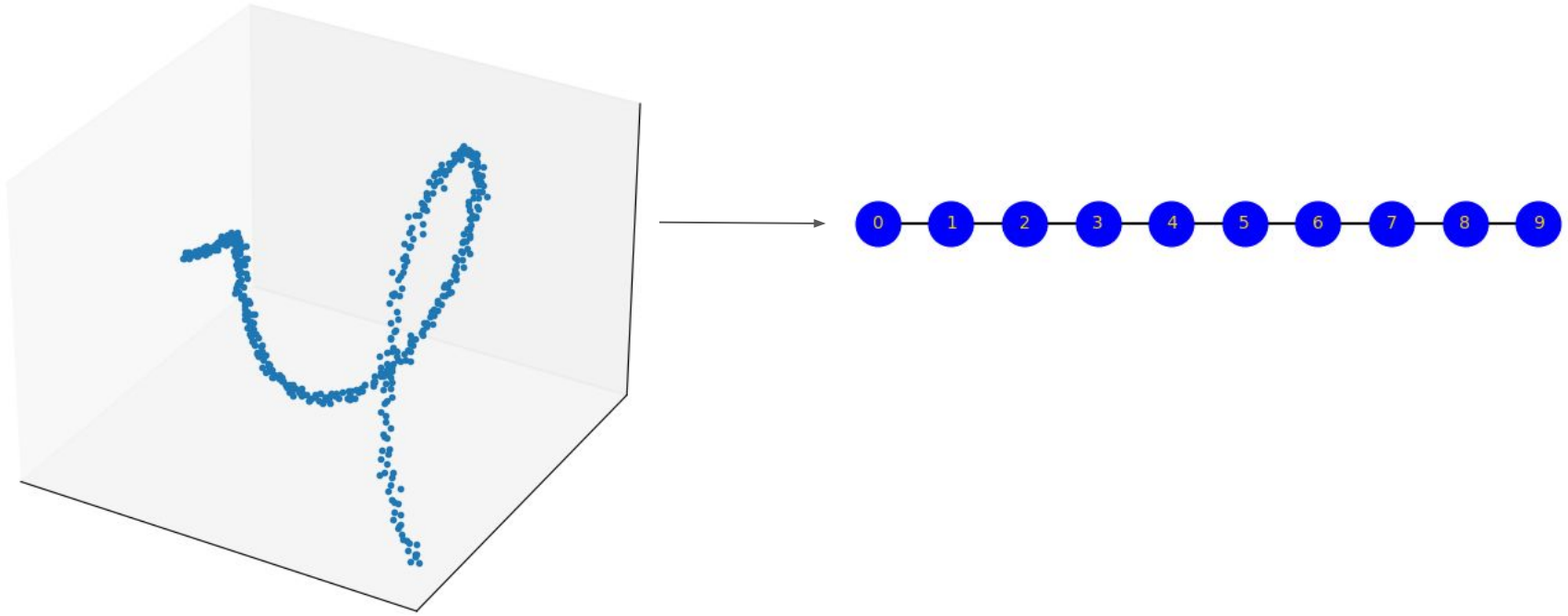
Wiskott, Laurenz. *Estimating Driving Forces of Nonstationary Time Series with Slow Feature Analysis*. no. December (2003): 1–8. <https://doi.org/10.1145/947121.947143>

Yan, Jinling, Shao-Wu Zhang, Chihao Zhang, Weitian Huang, Jifan Shi, and Luonan Chen. "Dynamical Causality under Invisible Confounders." arXiv:2408.05584. Preprint, arXiv, August 10, 2024. <https://doi.org/10.48550/arXiv.2408.05584>

Sauer, Timothy D. "Reconstruction of Shared Nonlinear Dynamics in a Network." *Physical Review Letters* 93, no. 19 (2004): 1–4. <https://doi.org/10.1103/PhysRevLett.93.198701>

Gilpin, William. "Recurrences Reveal Shared Causal Drivers of Complex Time Series." *Physical Review X* 15, no. 1 (2025). <https://doi.org/10.1103/PhysRevX.15.011005>.

Kohonen's Self organizing Map (SOM)



Kohonen, T. "Things You Haven't Heard about the Self-Organizing Map." *IEEE International Conference on Neural Networks*, IEEE, 1993, 1147–56. <https://doi.org/10.1109/ICNN.1993.298719>.

Kohonen's Self organizing Map (SOM)

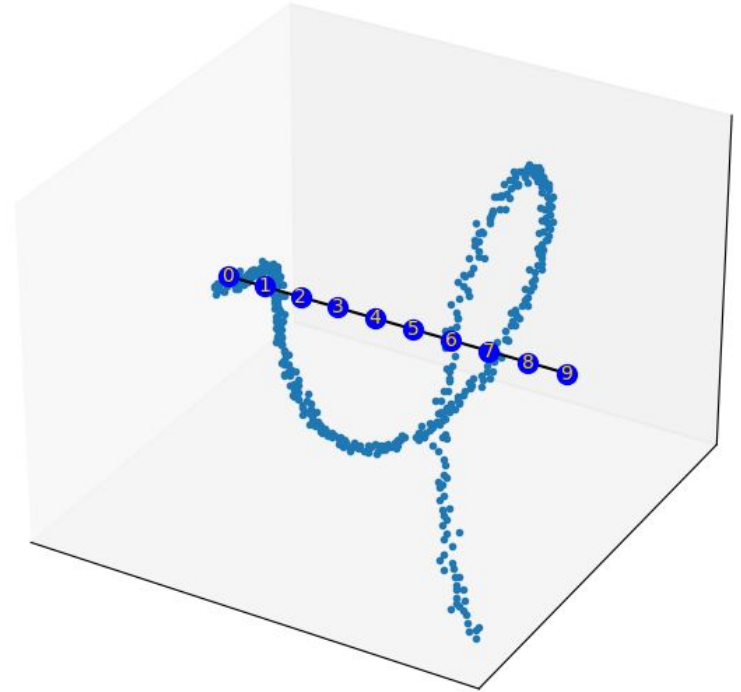
1. Sample an input $x(t)$
2. Compute distances $d_i(t) = ||x(t) - C_i(t)||$
3. Find BMU: $i^* = \operatorname{argmin}_i d_i(t)$
4. Compute neighborhood function:

$$h_{i,i^*}(t) = \exp\left(-\frac{(C_i - C_{i^*})^2}{2\sigma(t)^2}\right)$$

5. Update prototypes:

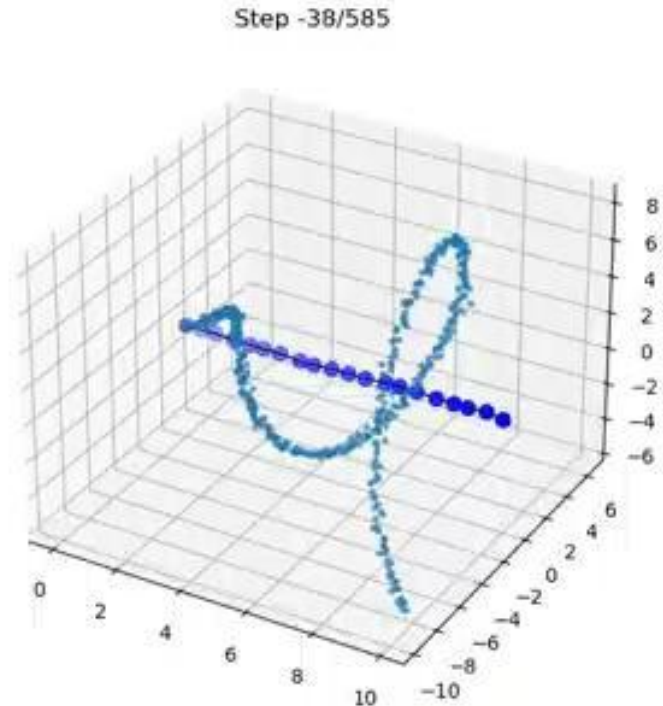
$$C_i(t+1) = C_i(t) + \varepsilon(t)h_{i,i^*}(t)[x(t) - C_i(t)]$$

6. Optionally decrease $\varepsilon(t)$ and $\sigma(t)$
7. Repeat for all training steps

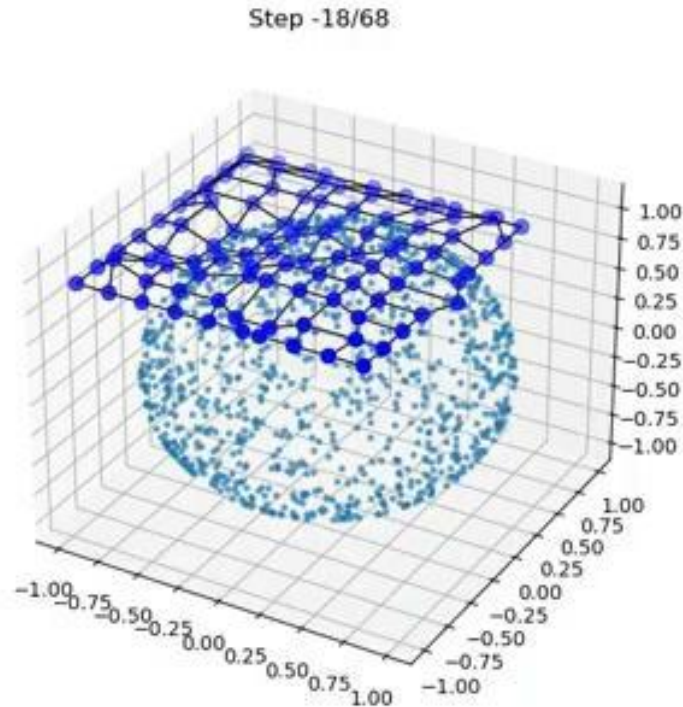


Kohonen's Self organizing Map (SOM)

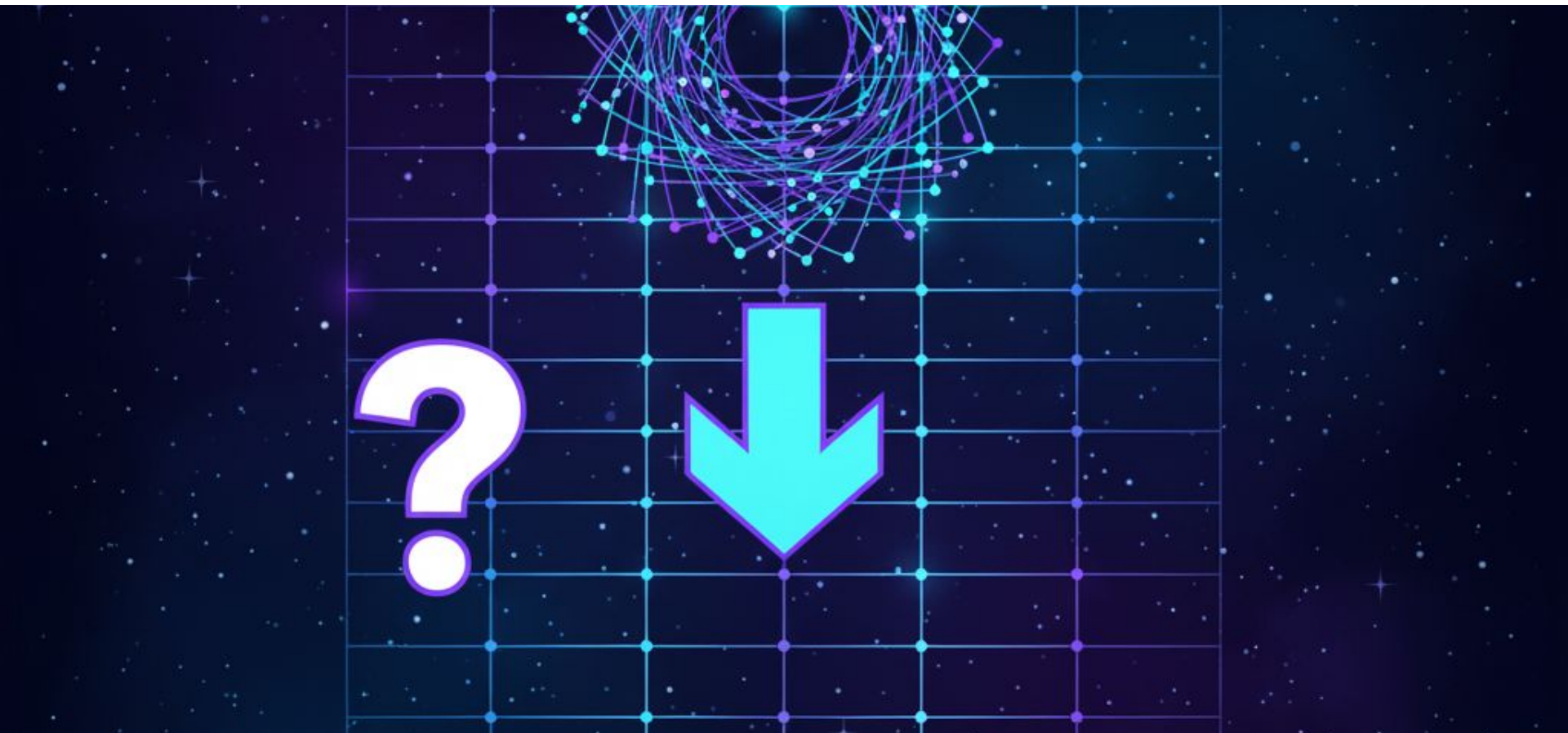
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5. Update prototypes:
$$C_i(t+1) = C_i(t) + \varepsilon(t)h_{i,i^*}(t)[x(t) - C_i(t)]$$
6. Optionally decrease $\varepsilon(t)$ and $\sigma(t)$
7. Repeat for all training steps



Kohonen's Self organizing Map (SOM)



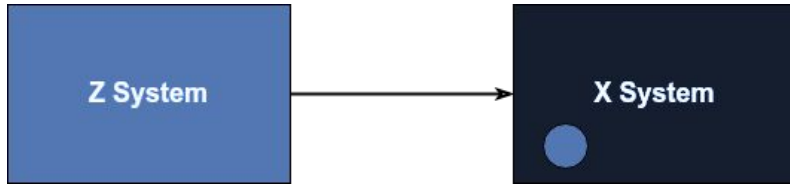
How to introduce inductive bias?



Theoretical & Practical interlude #X – Adventures in
reconstructed state-spaces 2

Coupled dynamical systems and shared dynamics

Unidirectional coupling

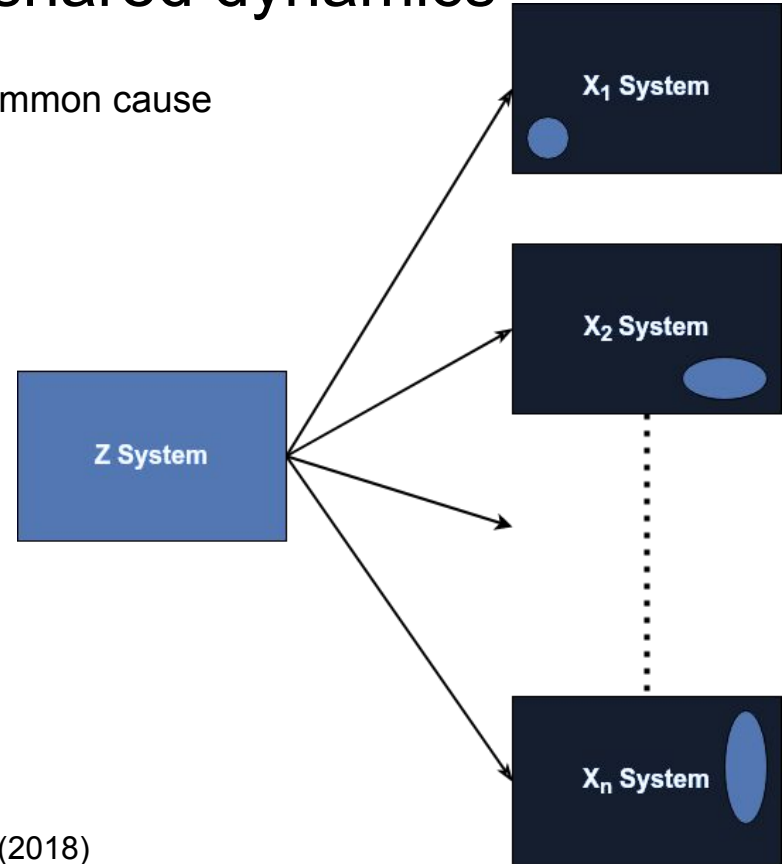


Causal effect leaves a mark on the dynamics

A common cause means shared shared dynamics in the effects

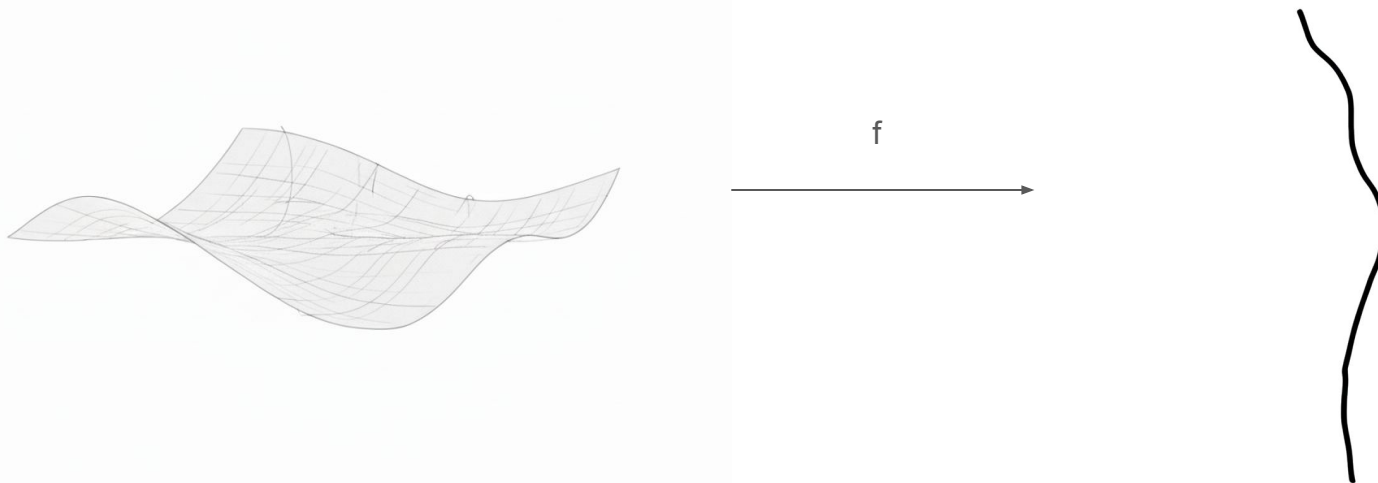
Possibility of decomposition!

Common cause



Preimage theorem

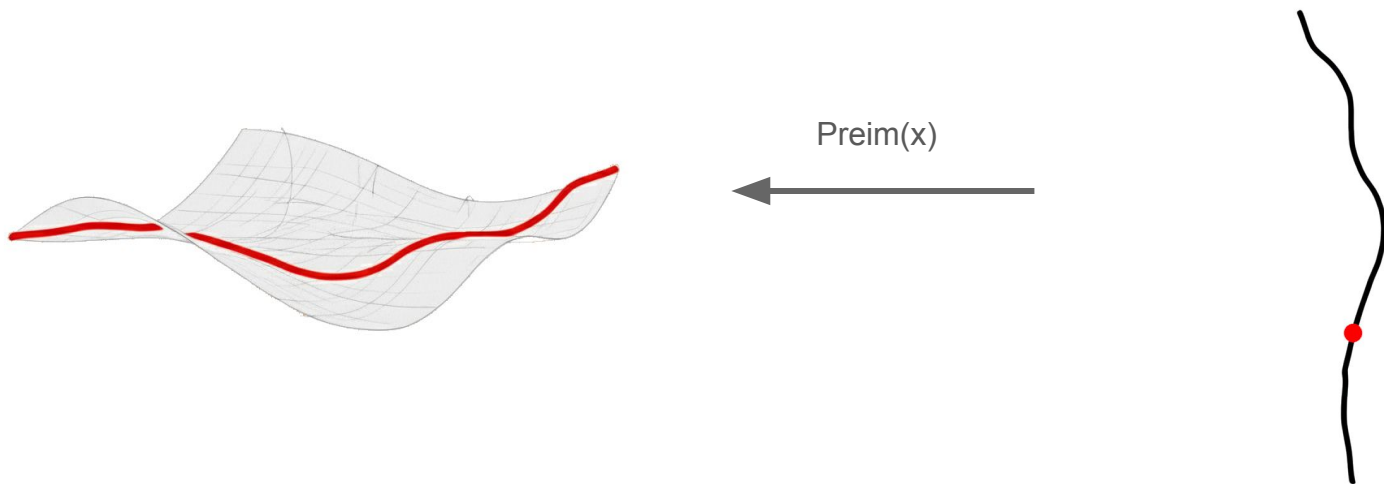
Let Y and X be Euclidean manifolds with dimensions D and d respectively ($D > d$) and let $f : Y \rightarrow X$ be a smooth mapping and x in X a regular value of the mapping. Then, the preimage $f^{-1}(x)$ is a submanifold of Y with codimension $\dim(x)$.



- + Sard theorem: regular points are dense (the Lebesgue measure of irregular points is zero) for a generic smooth mapping

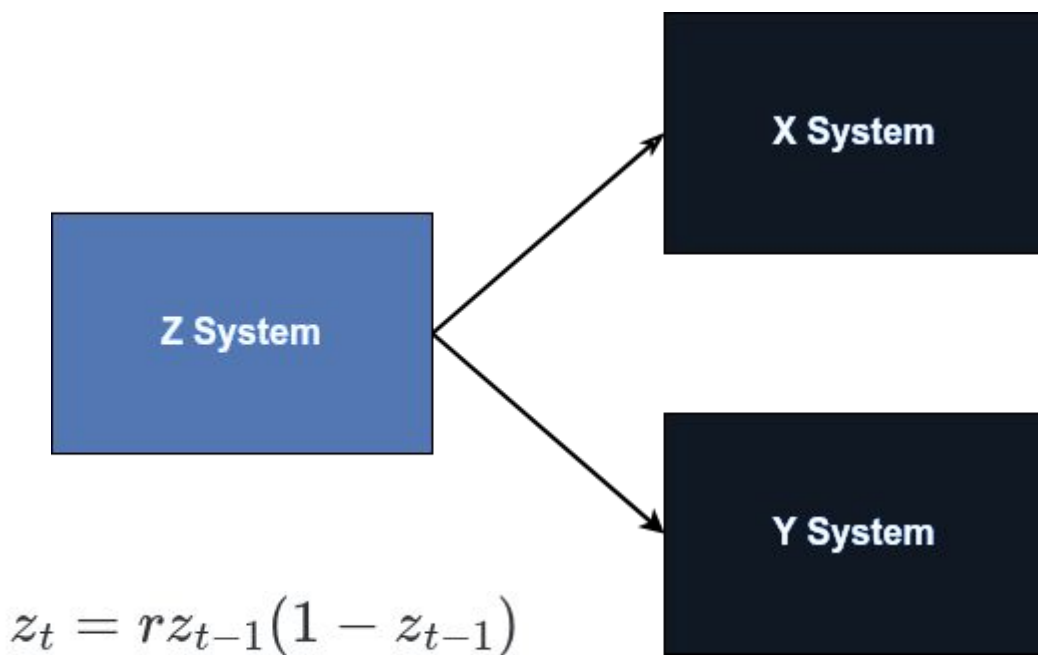
Preimage theorem

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- + Sard theorem: regular points are dense (the Lebesgue measure of irregular points is zero) for a generic smooth mapping

Example 1 - Logistic maps



Unobserved

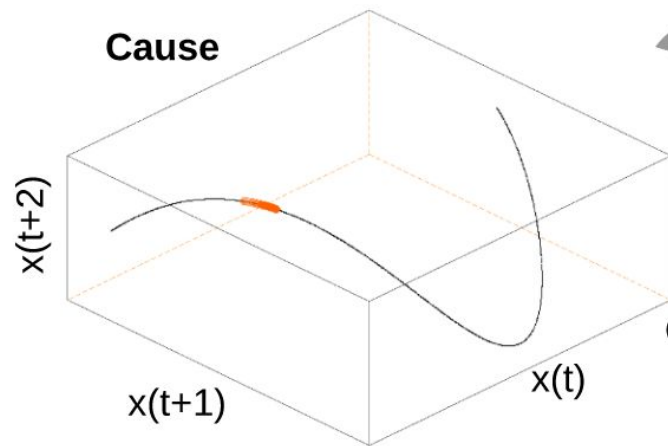
Observed

$$x_t = rx_{t-1}(1 - x_{t-1} - \beta_{xz}z_{t-1})$$

$$y_t = ry_{t-1}(1 - y_{t-1} - \beta_{yz}z_{t-1})$$

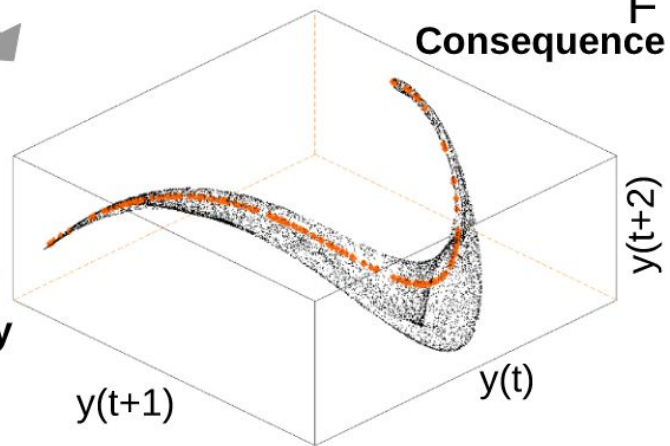
Goal: Reconstruct Z!

E

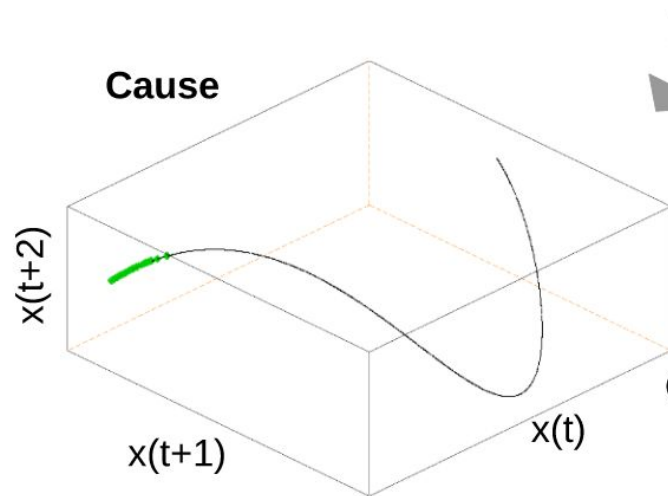


Mapping

Causality

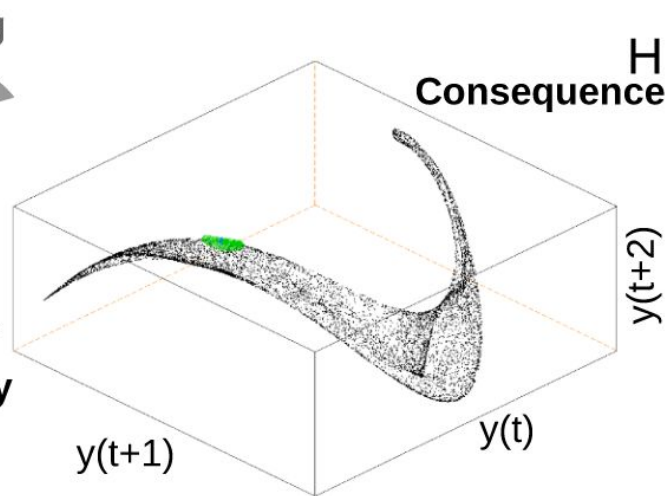


G

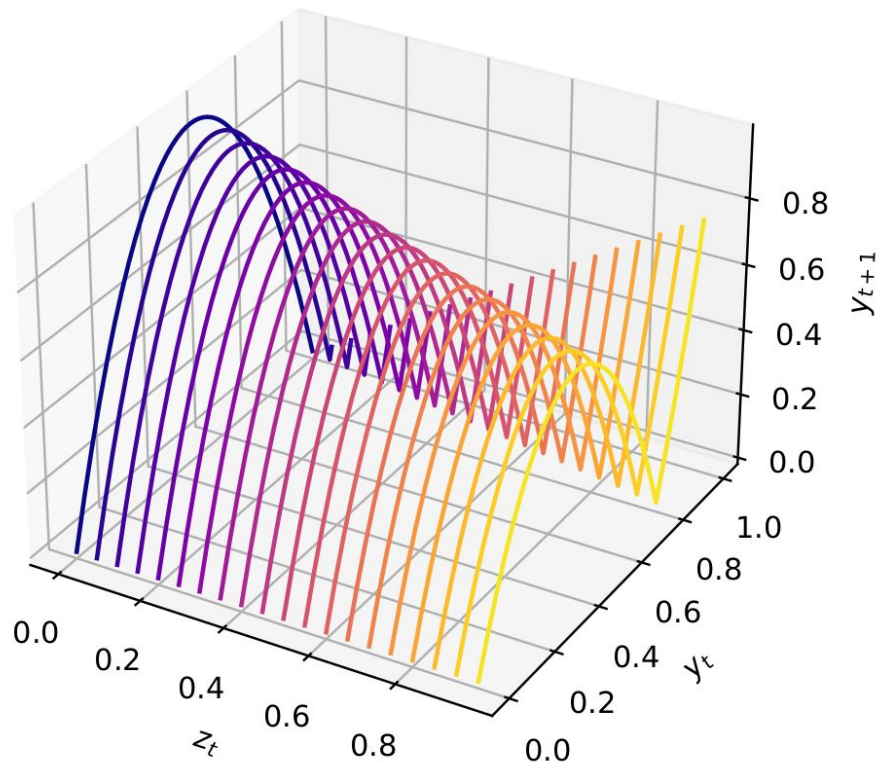


Mapping

Causality



Mapping from Y to Z



$$f : \quad x_t = rx_{t-1}(1 - x_{t-1} - \beta_{xz}z_{t-1})$$

$$g : \quad y_t = ry_{t-1}(1 - y_{t-1} - \beta_{yz}z_{t-1})$$

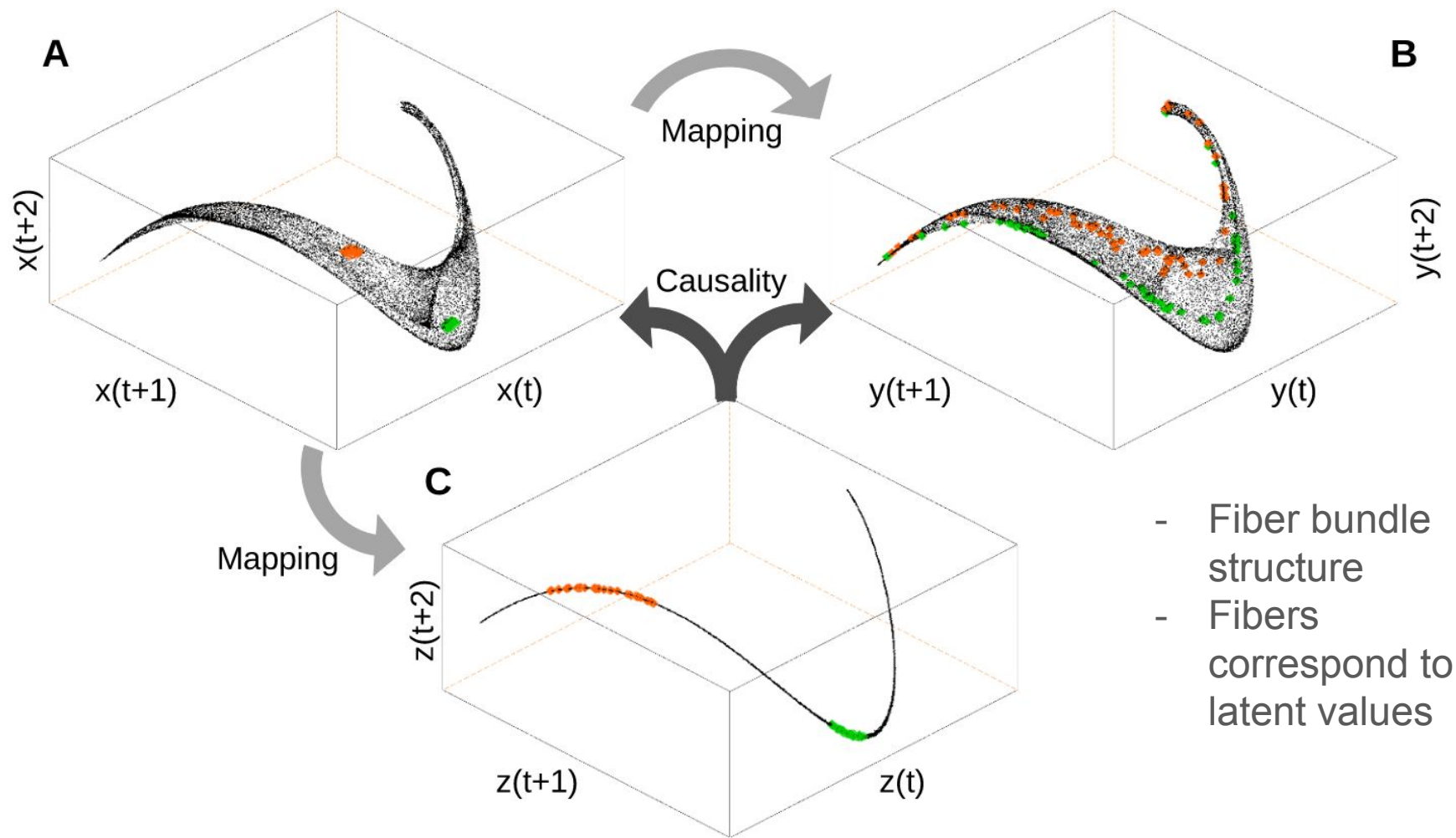
$$h : \quad z_t = rz_{t-1}(1 - z_{t-1})$$

$$Y_t = [y_t, y_{t-1}] \longrightarrow S_t = [y_{t-1}, z_{t-1}]$$

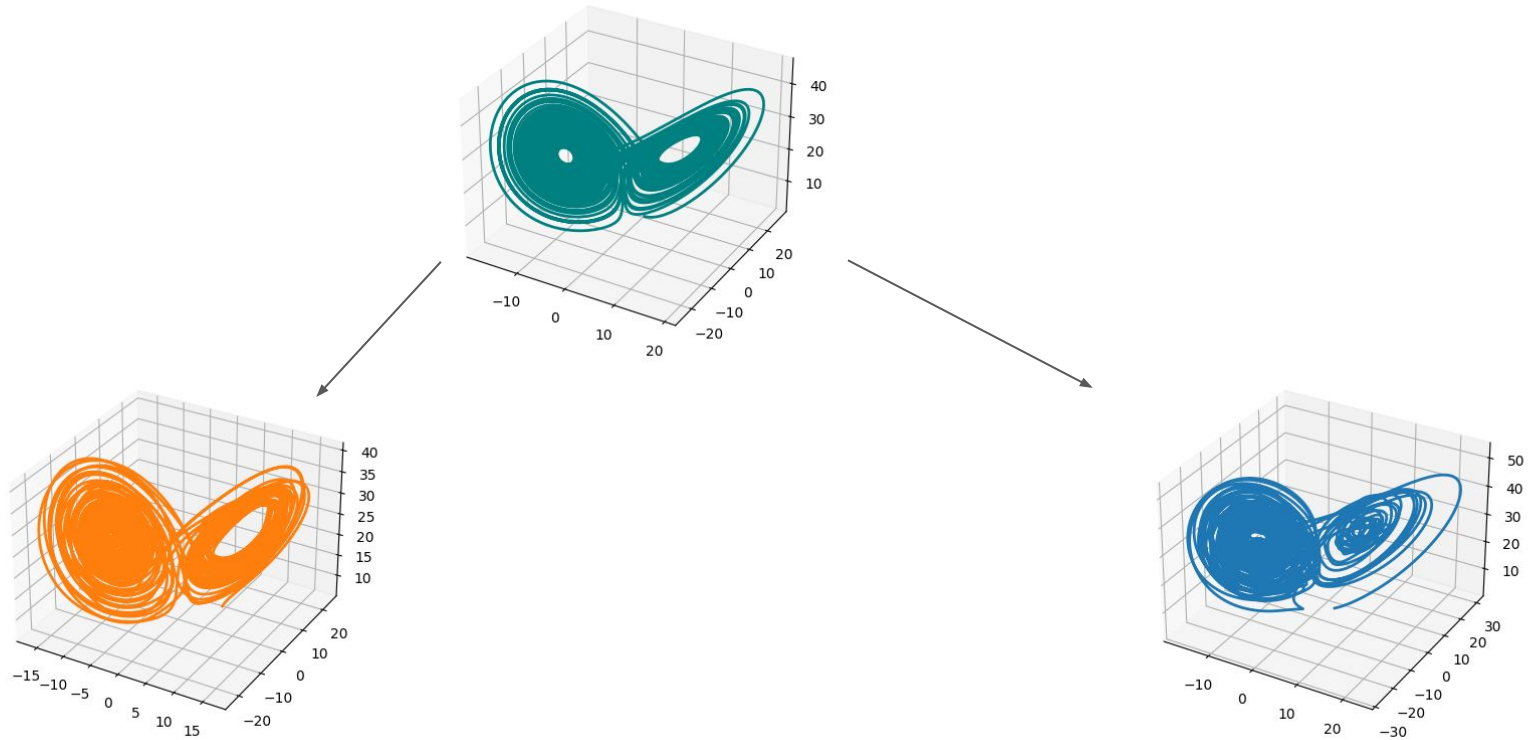
$$\phi : \quad z_{t-1} = \frac{1 - y_{t-1} - \frac{y_t}{ry_{t-1}}}{\beta_{yz}}$$

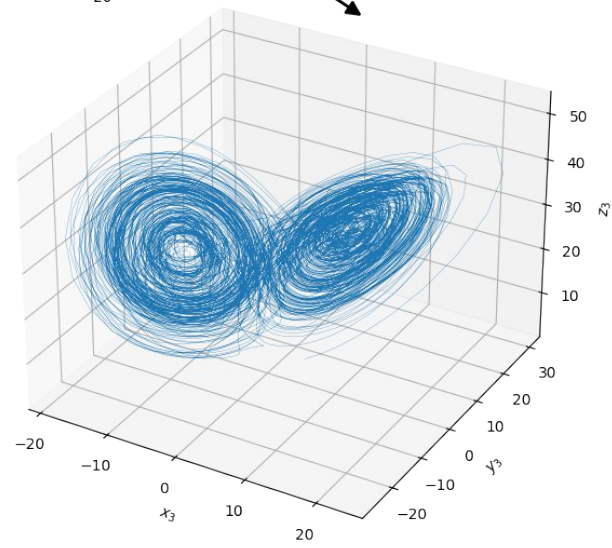
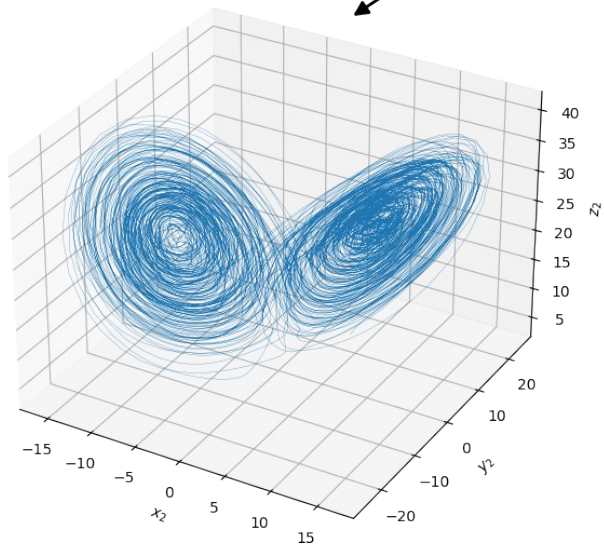
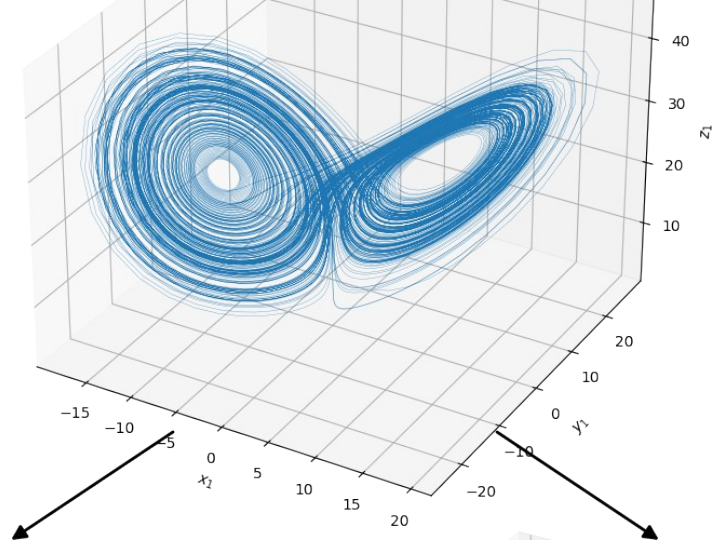
Embedding of observed dynamics $X(t)$

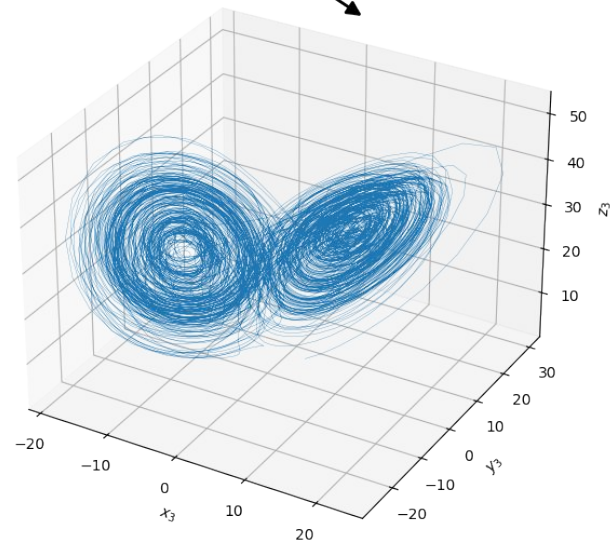
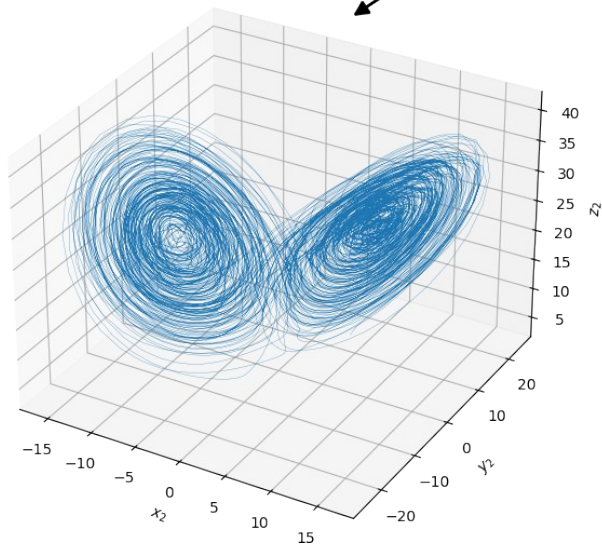
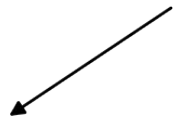
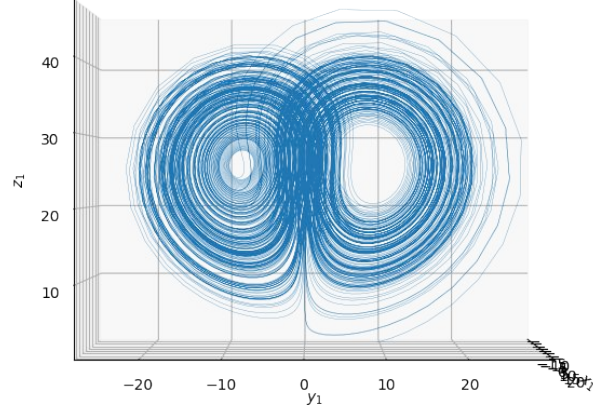
Embedding of observed dynamics $Y(t)$

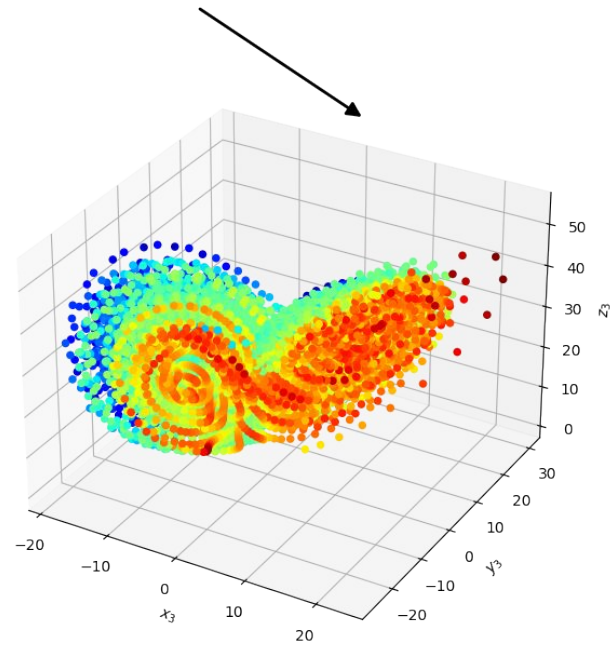
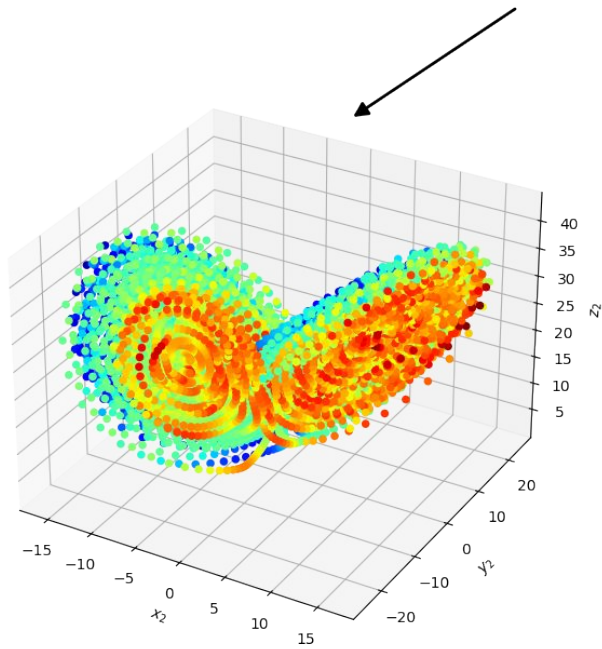
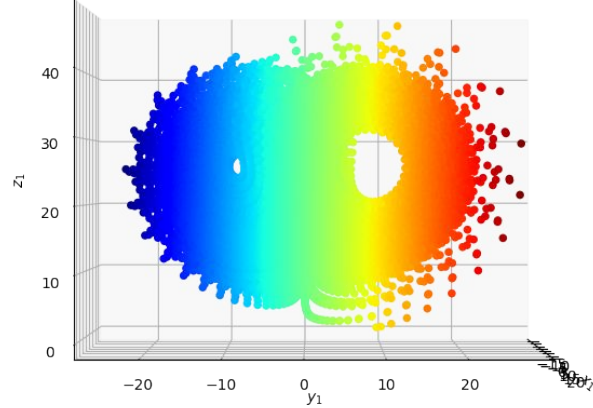


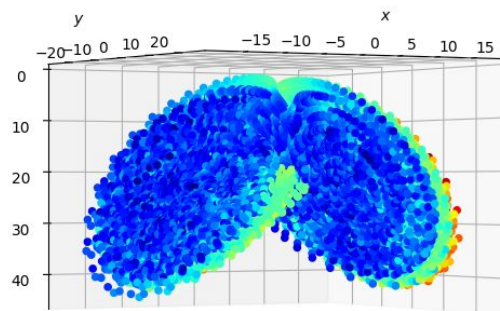
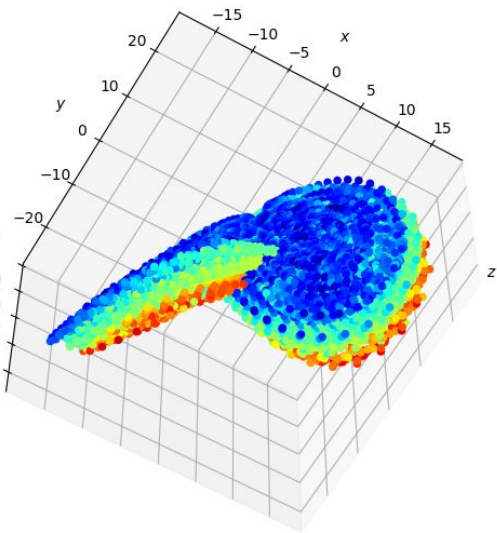
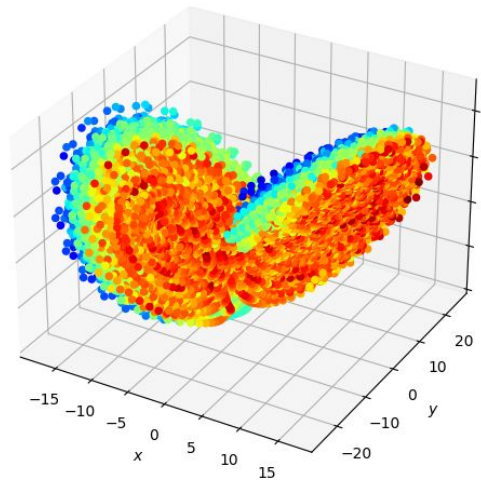
Example 2 - Lorenz system











Anisotropic Self-Organizing Map

Algorithm 1 Training algorithm for ASOM.

$$X(t) \leftarrow [x(t), x(t + \tau), \dots, x(t + (m - 1)\tau)]$$

$$Y(t) \leftarrow [y(t), y(t + \tau), \dots, y(t + (m - 1)\tau)]$$

$$\gamma \leftarrow 0$$

$$C_{i,j}(\gamma) \leftarrow r_{ij}$$

for $s = 0$ to N **do**

 Compute $\sigma_1(s), \sigma_2(s), \epsilon(s)$

$t_s \leftarrow r_s$

$i^*, j^* \leftarrow \arg \min_{i,j} (||C_{i,j}(\gamma) - Y(t_s)||)$

$t_1, \dots, t_K \leftarrow \text{KNN}(X(t_s))$

for $k = 1$ to K **do**

$i^a \leftarrow \arg \min_i ||C_{i,j^*}(\gamma) - Y(t_k)||$

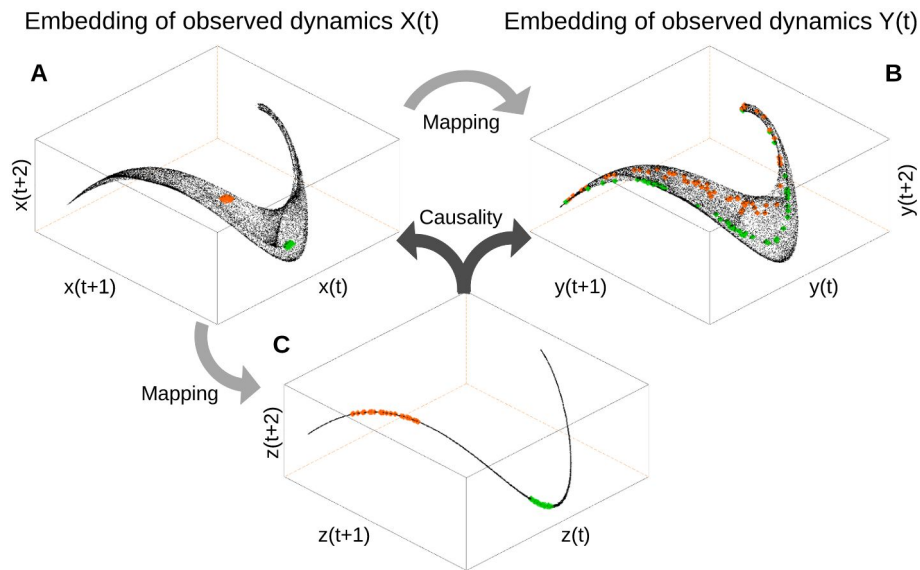
$w_{i,j}(\gamma + 1) \leftarrow \exp [-((i - i^a)^2 / \sigma_1^2(\gamma) + (j - j^*)^2 / \sigma_2^2(\gamma))]$

$C_{i,j}(\gamma + 1) \leftarrow C_{i,j}(\gamma) + \epsilon(s)w_{i,j}(\gamma)(Y(t_k) - C_{i,j}(\gamma))$

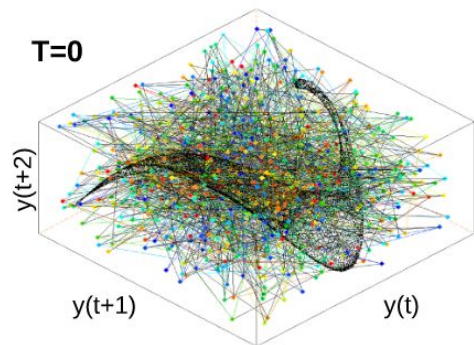
$\gamma \leftarrow \gamma + 1$

end for

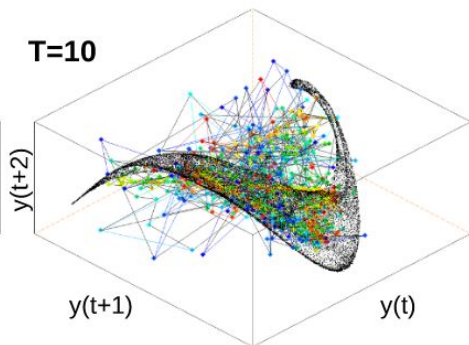
end for



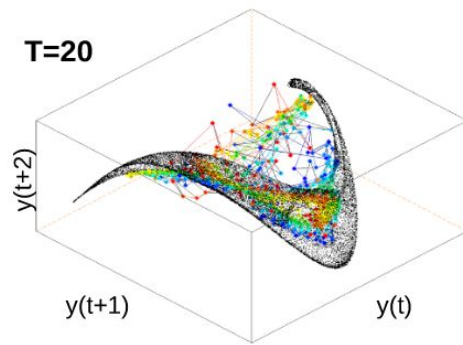
T=0



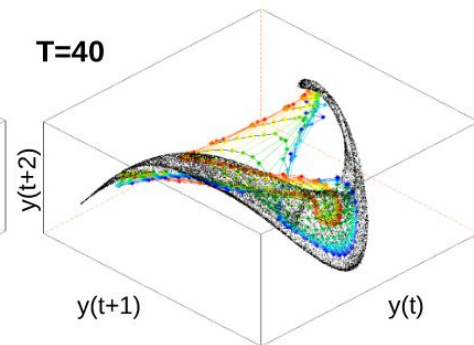
T=10



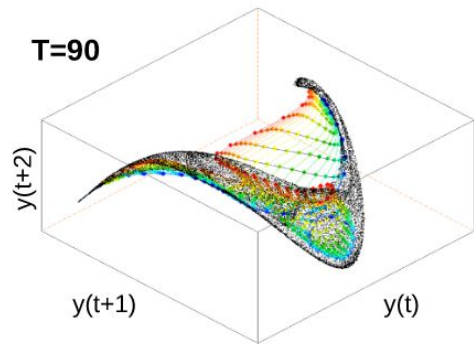
T=20



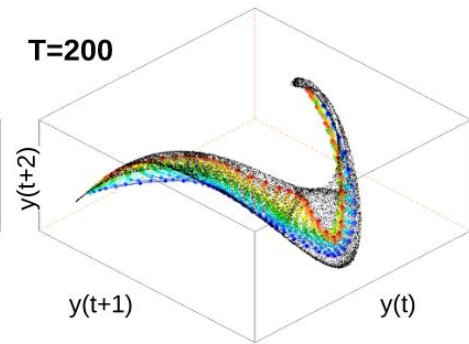
T=40



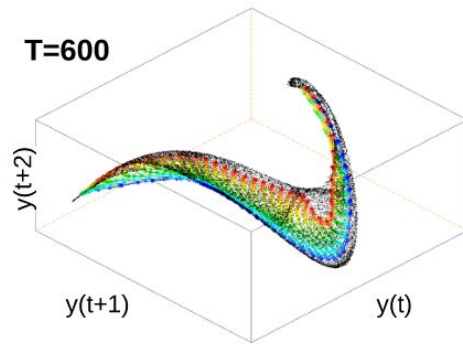
T=90



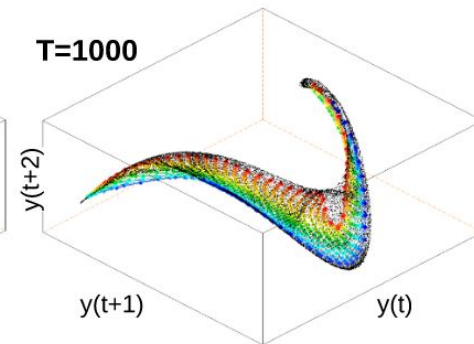
T=200



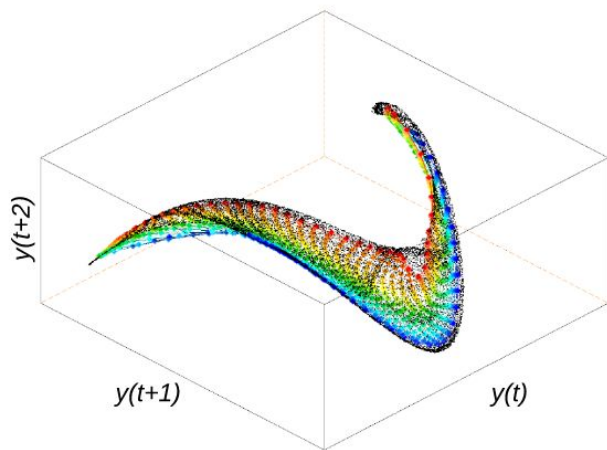
T=600



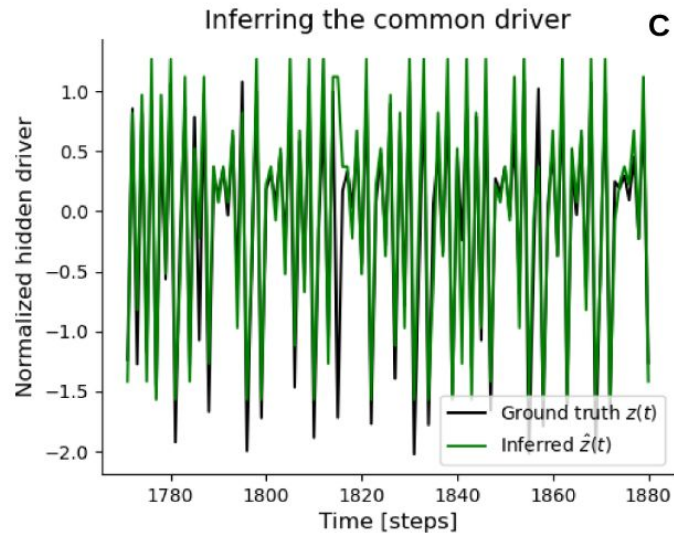
T=1000



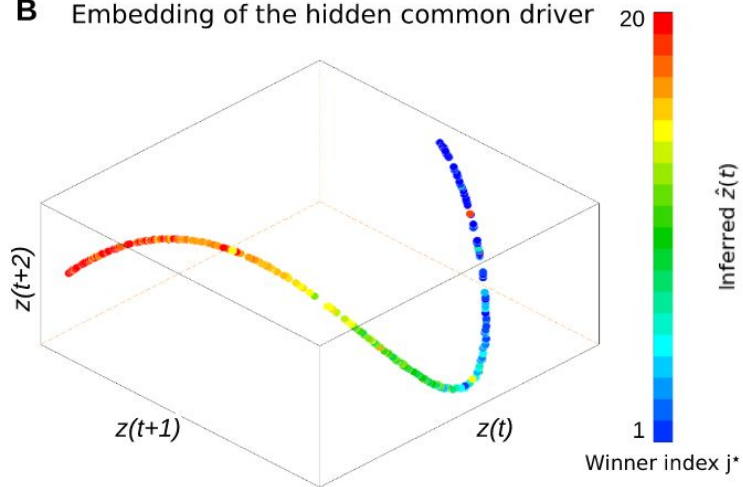
A Learned topology by ASOM, $T=10000$



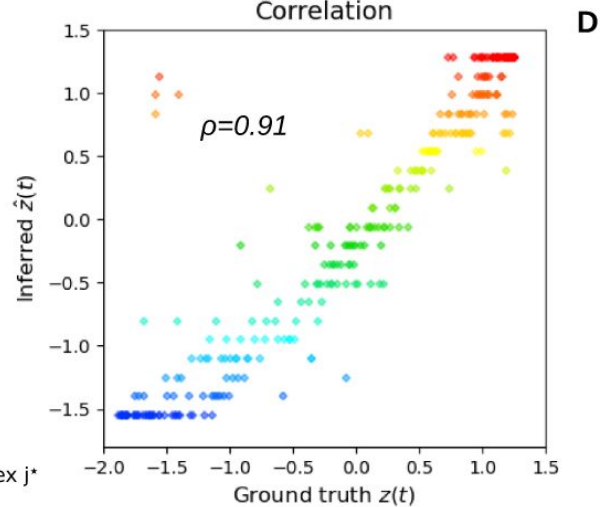
Inferring the common driver



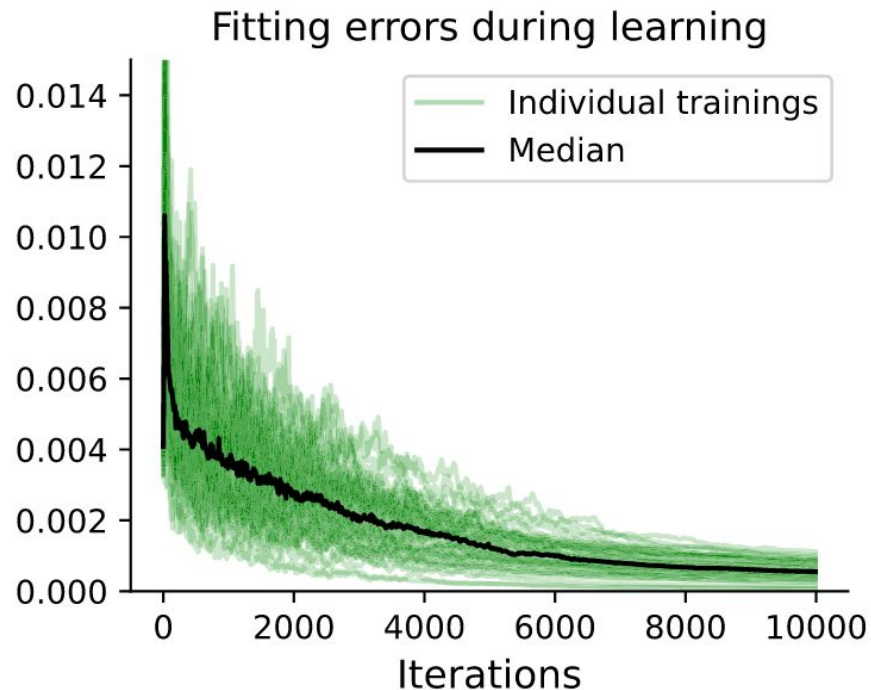
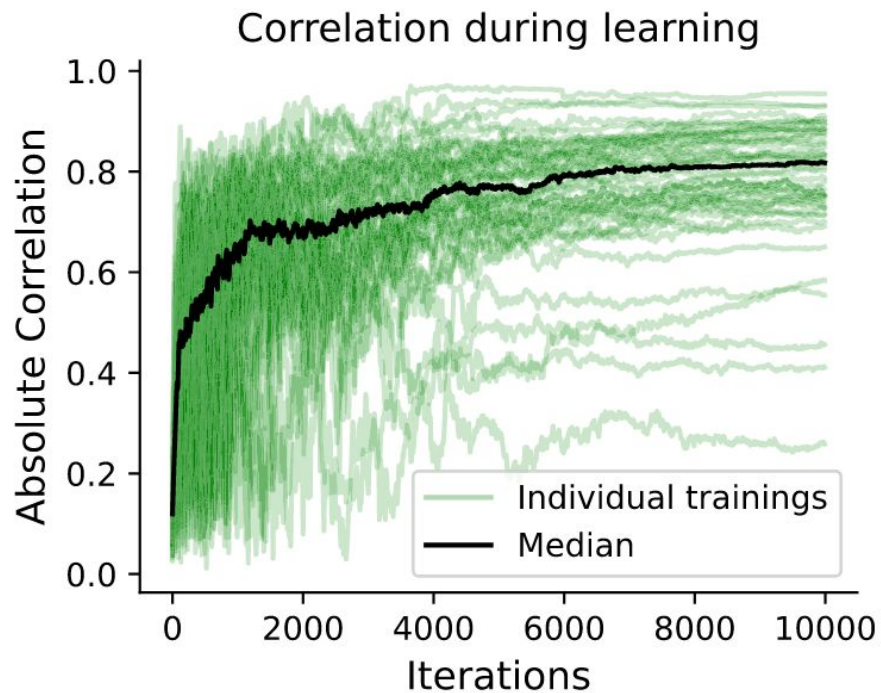
B Embedding of the hidden common driver



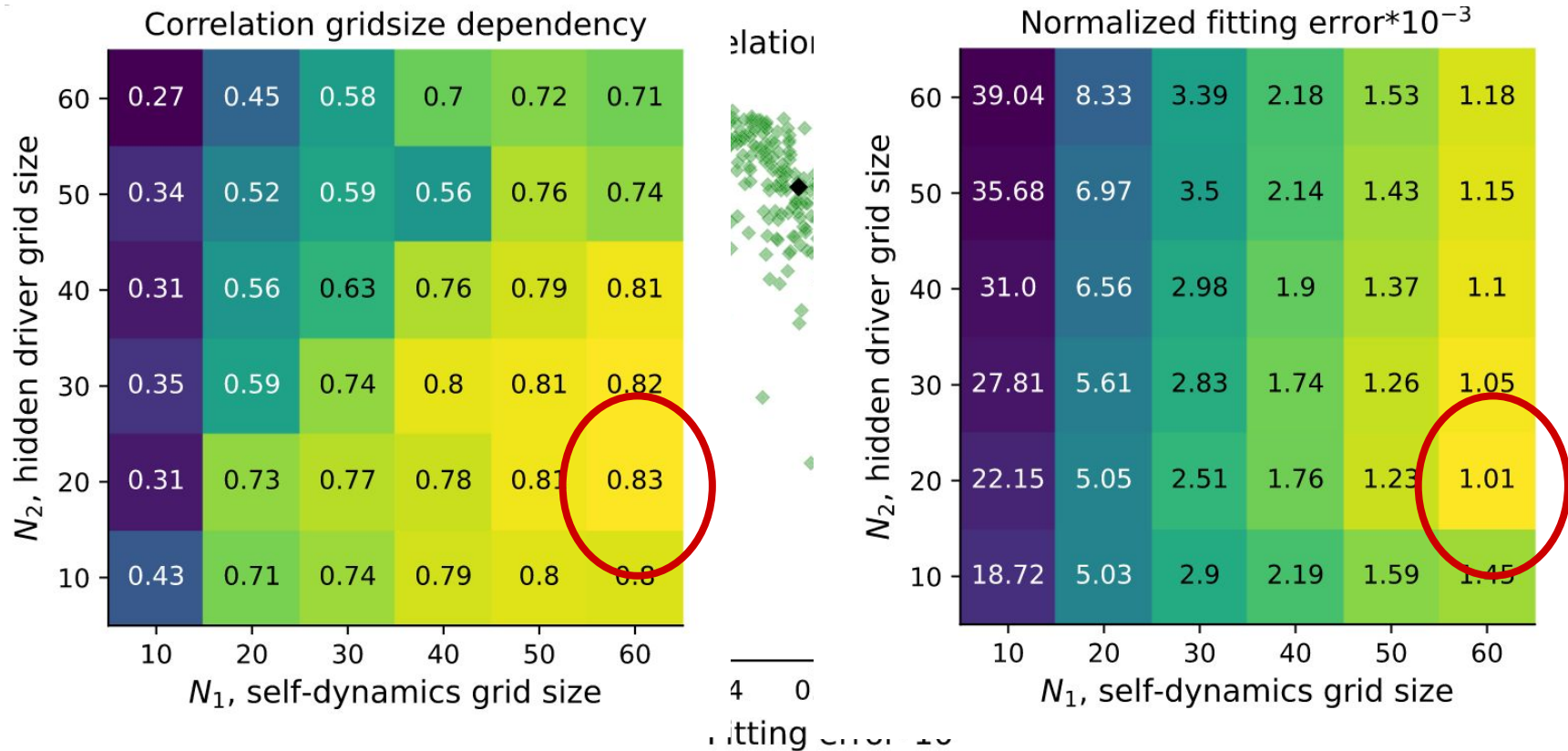
Correlation



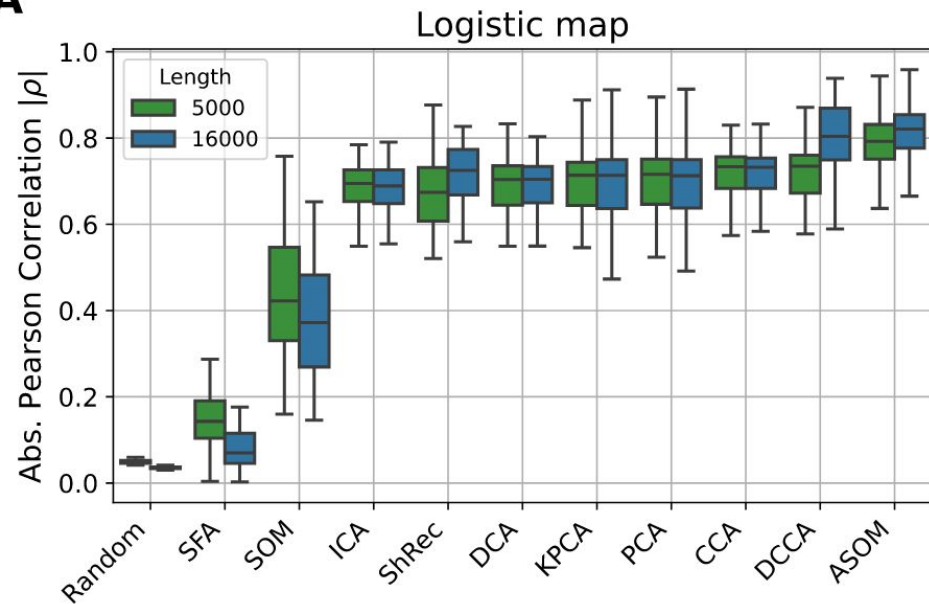
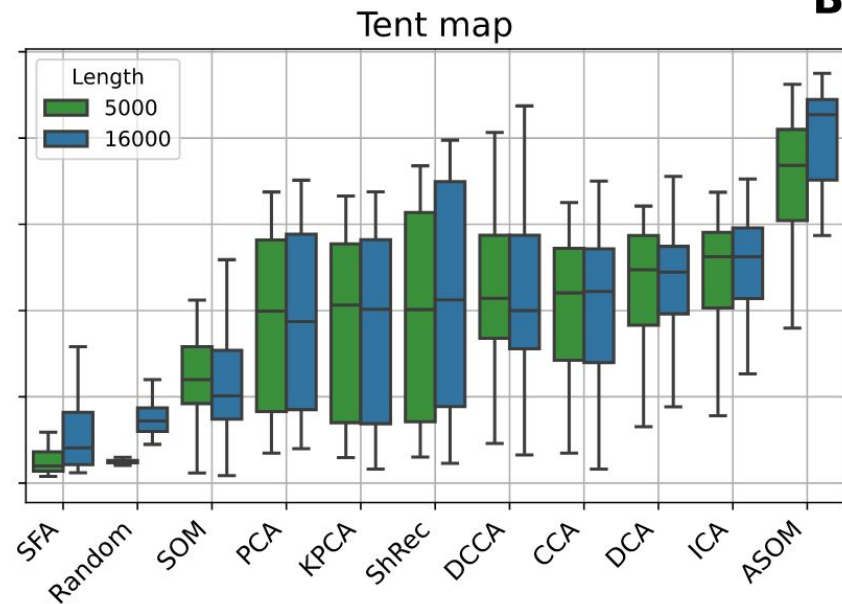
Performance & fitting error



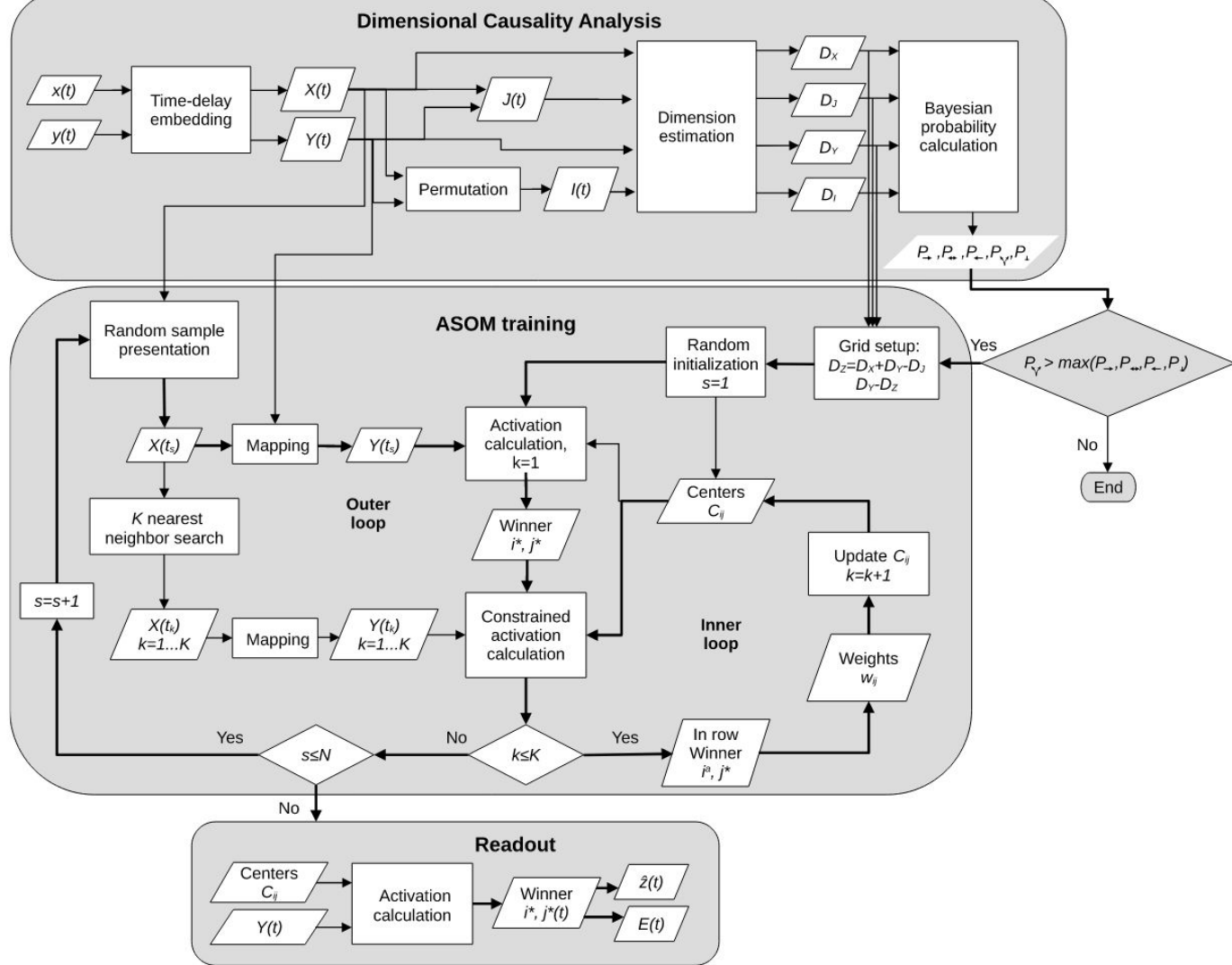
Fitting error: Proxy for hyperparameter tuning



Comparison

A**B**

The process:



Summary

- ASOM for hidden common driver reconstruction
- Superior performance on simulated examples (PCA, SFA...)
- Some challenges:
 - Fitting of hyperparameters (normalized fitting error)
 - Extension to higher dimensions (Benko, 2018)
- Future directions:
 - Higher order decomposition
 - Other architectures

Thank you!



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